

Separating drug main effects from molecular personalization in cancer drug-sensitivity prediction: a harmonized GDSC/CTRP and CCLE benchmark with permutation-controlled biomarker discovery

Yue Hao^{1*}

¹ Independent Researcher, Chandler AZ, 85248 USA

* Corresponding author: yueyvett hao@gmail.com; ORCID: 0000-0002-5065-8906

Abstract

Large pharmacogenomic screens promise to predict which drugs a tumour will respond to from its molecular profile, yet it remains unclear how much of the apparent predictive skill, that is, performance in excess of a baseline that ignores the molecular profile entirely, reflects genuine molecular personalization rather than simpler structure in the data. Here I integrate two independent drug-sensitivity screens, the Genomics of Drug Sensitivity in Cancer (GDSC) and the Cancer Therapeutics Response Portal (CTRP), with multi-omic profiles from the Cancer Cell Line Encyclopedia (CCLE), resolving cell-line identity through Cancer Dependency Map (DepMap) identifiers and drug identity through chemical structure. The harmonized resource spans 678 drugs across 1,171 cell lines (487,799 measurements), with a cross-cohort-comparable sensitivity metric validated against known biology. Using grouped, leave-cell-line-out nested cross-validation scored against a drug-mean baseline, I decompose prediction into a broad-spectrum component (which drugs are generically potent within a tissue) and a mechanism-specific component (personalization surviving after drug means are removed). The broad-spectrum component is strong within every tissue, which identifies it as tissue-by-drug structure rather than molecular personalization; it necessarily vanishes when tissues are pooled, because the metric is standardized per drug across the whole panel, so the pooled comparison confirms the interpretation rather than establishing it. Pooling all tissues does expose a small personalization signal that survives removal of tissue-of-origin structure (held-out Spearman $\rho \approx 0.05$ against a within-tissue permutation null), though held-out skill falls by roughly two-fifths once tissue is removed, a tissue-only baseline outscores the molecular model on the same target, and the null constrains lineage rather than every line-level confound. For per-drug discovery I introduce a two-stage significance protocol, a permutation null followed by independent corroboration, and apply it across all 7,836 drug-tissue panels: the calibrated null retains roughly a threefold smaller set than cross-validation confidence intervals admit, leaving candidate biomarkers for a minority (~12%) of drug-tissue pairs at a false-discovery proportion of roughly 0.4, a set that is enriched over fourfold for established drug-target pharmacology and that surfaces novel, recurrent vulnerabilities, most strikingly a melanoma ferroptosis signal shared by three independent GPX4 inhibitors. Taken together, these results support baseline-anchored, permutation-controlled evaluation and drug recommendation that is tiered by confidence.

Keywords: drug sensitivity prediction; pharmacogenomics; cancer cell lines; data harmonization; cross-validation; permutation testing

1. Introduction

Panels of molecularly characterized cancer cell lines paired with high-throughput drug screening underpin much of modern preclinical pharmacogenomics. The Genomics of Drug Sensitivity in

Cancer (GDSC) project systematically screened hundreds of compounds across nearly a thousand lines and related sensitivity to genomic markers [1, 2, 3]. In parallel, the Cancer Therapeutics Response Portal (CTRP) profiled an “informer set” of small molecules to connect sensitivity to cellular features and mechanism of action [4, 5, 6]. Both build on the Cancer Cell Line Encyclopedia (CCLE), which provides deep multi-omic characterization (expression, mutations, copy number, and more) of the same lines [7, 8], now maintained and extended within the Cancer Dependency Map (DepMap) [9]. Together these resources have become the default substrate for building and benchmarking models that predict drug response from molecular profiles.

The central aspiration is *personalized* prediction: given a tumour’s molecular state, rank the drugs to which it is unusually vulnerable. A large methodological literature, spanning regularized linear models to deep neural networks, now addresses this task, catalyzed by community efforts such as the DREAM drug-sensitivity prediction challenge [10] and summarized in several reviews [11, 12]. Yet progress has been uneven, and a recurring observation is that sophisticated models often fail to convincingly beat simple baselines, and that performance degrades sharply when models are transferred across studies rather than evaluated within a single dataset [13]. This observation also dictates the model class used here. A systematic, multifactorial assessment of modeling choices on these same cell-line panels found that which compound is being predicted and which molecular features are supplied dominate predictive performance, while the choice of learning algorithm contributes comparatively little: penalized linear methods such as the elastic net matched or exceeded more expressive nonlinear alternatives [14]. In a regime of tens of thousands of correlated molecular features measured on at most a few hundred cell lines per tissue, added model capacity therefore fits noise rather than improving out-of-sample prediction, so I use elastic-net regression for every multivariate model here and let the comparisons turn on features, baselines, and evaluation discipline rather than on architecture; the per-drug discovery analysis of Section 2.6 is deliberately simpler still, a supervised correlation ranking rather than a fitted regression.

Two well-documented confounds help explain the gap between personalized prediction and drug screening data, and both are central to the present work. First, the two major screens do not agree as strongly as one might hope: cross-study concordance of drug-response measurements is only moderate, which caps the reproducibility of any biomarker learned from one screen and tested on another [15]. Second, and more subtly, much of the “signal” in these panels is not molecular personalization at all but *drug main effects*: some compounds are broadly potent across most cell lines, and broad tissues differ in their general level of drug sensitivity. Accounting for these general sensitivity levels has been shown to change, and often improve, biomarker discovery [16], and baseline gene-expression models can appear predictive largely by recapitulating tissue-of-origin structure [17]. A model that predicts “this drug is generally potent” or “this lineage is generally sensitive” can post respectable correlations while personalizing nothing.

These confounds motivate an evaluation discipline that is intentionally stricter than what is commonly reported. If a molecular model cannot beat a baseline that predicts each drug’s average sensitivity while *ignoring* the tumour’s profile, then it is re-learning drug main effects, not personalizing. If a claimed biomarker is significant only under a confidence interval that measures cross-validation split-stability rather than sampling noise, it may not survive a proper null. Both failure modes are easy to commit and hard to see without deliberately testing for them.

A harmonized GDSCUCTRP drug-response resource was built and joined to CCLE multi-omics, resolving cell lines to DepMap identifiers and resolving drug identity on chemical structure rather than name, so that the same molecule screened under two aliases pools correctly while two different molecules sharing a name never do. I then evaluate drug-sensitivity prediction under grouped, leave-

cell-line-out nested cross-validation, always scored against a drug-mean baseline, and explicitly decompose the problem into a broad-spectrum component and a mechanism-specific component (the personalization signal that survives after drug main effects, and, when tissues are pooled, tissue of origin, are removed). I test whether raw expression, multi-omic layers, or Reactome pathway activity add mechanism signal, and I use permutation testing to ask how many per-drug biomarkers are truly significant versus artifacts of an over-optimistic confidence interval. The result is not a new state-of-the-art predictor but an honest map of *where* predictive signal genuinely lives in these data (broad-spectrum within tissue, and mechanism-specific for a discoverable minority of targeted agents), together with a cautionary quantification of how conventional evaluation inflates discovery.

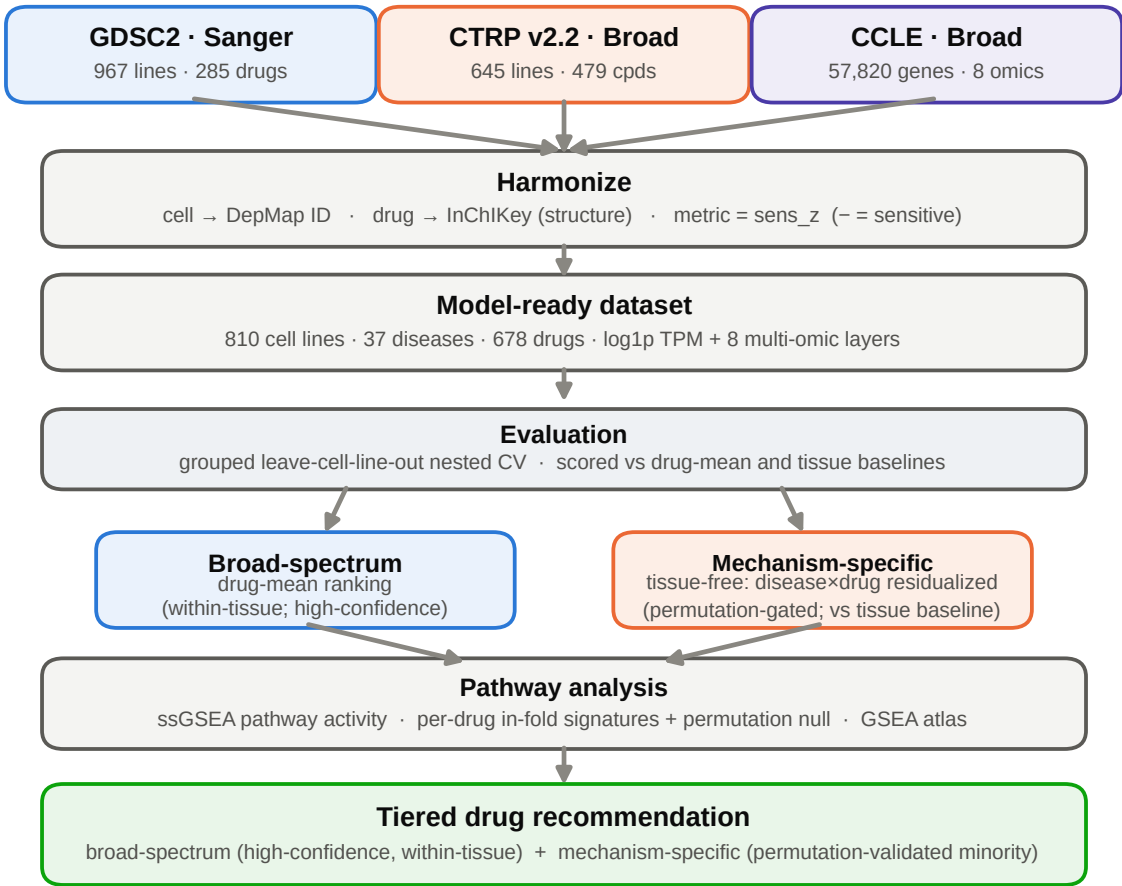


Figure 1. Overview of the harmonized GDSC-CTRP-CCLE resource and the baseline-anchored evaluation pipeline, from raw screens through structure-resolved harmonization to the tiered, permutation-gated recommendation. The mechanism-specific arm is evaluated tissue-free: within each training fold the disease-by-drug (tissue×drug) cell mean is subtracted from the target so tissue of origin is removed before the expression model is fit, and the arm is scored against both a drug-mean baseline and a no-feature tissue baseline (the in-fold tissue×drug mean) under a within-tissue permutation null.

2. Materials and Methods

2.1 Data sources

Three public resources were used, all in their released form. Drug-response data came from GDSC2 (the second-generation GDSC screen; fitted dose-response release of 27 Oct 2023) [1, 2, 3] and from CTRP v2.2 [4, 5, 6]. GDSC2 reports, per drug $\times$ cell line, the natural-log half-maximal inhibitory concentration (LN IC50), a normalized area under the dose-response curve (AUC, bounded 0–1), and a per-drug Z-score. Molecular data came from the CCLE 2019 release [7, 8]. The primary feature space was RNA-seq expression (RSEM transcripts-per-million, TPM, with raw counts retained for the differential-expression analyses), and eight companion assays were used as add-on layers: somatic mutation calls, gene fusions (CCLE_Fusions_20181130), reduced-representation bisulphite sequencing methylation summarized over 1 kb windows around transcription start sites (CCLE_RRBS_TSS1kb_20181022), targeted metabolomics (CCLE_metabolomics_20190502), reverse-phase protein array (RPPA) with its antibody-to-gene annotation (CCLE_RPPA_20181003), microRNA expression indexed by mature-miRNA accession (CCLE_miRNA_MIMAT), global chromatin profiling of histone-mark combinations (CCLE_GlobalChromatinProfiling_20181130), and ABSOLUTE-derived ploidy and whole-genome-doubling calls (CCLE_ABSOLUTE_combined_20181227). Cell-line annotations and identifiers were taken from the CCLE/DepMap annotation file (Cell_lines_annotations_20181226) [9].

2.2 Cell-line and drug harmonization

Cell lines were resolved to a single canonical key, the DepMap identifier; unmatched Sanger/GDSC/CTRP lines were retained under explicit fallback keys but lack CCLE omics downstream. **Drugs** were resolved on chemical structure rather than name. CTRP compound SMILES notations were converted to InChI/InChIKeys with RDKit 2026.3.3 [18, 19]; GDSC compounds, which ship without structures in the raw release, were resolved from drug names and synonyms via PubChem [20]. Cross-cohort drug identity was then defined by the InChIKey connectivity block (the first 14 characters), which is tolerant of stereochemistry and salt form. Crucially, compounds sharing a name but differing in connectivity block (e.g., the bosutinib isomer) were flagged and never pooled. Resolving on structure rather than name increased the number of confidently shared compounds from 71 (naive name overlap) to 86.

2.3 A cross-cohort-comparable sensitivity metric

Because LN IC50 is available only for GDSC and the two AUC definitions are not comparable, sensitivity was expressed as a within-cohort, within-drug standardized score. For each (cohort, drug) group, the raw AUC was converted to a z-score, **sens_z**, removing each drug’s potency offset and each cohort’s scale/definition offset simultaneously. By construction, lower AUC corresponds to greater drug effect, so **more negative sens_z denotes greater sensitivity** in both cohorts; a distribution-free per-group percentile rank was retained as a companion. Raw AUC, LN IC50, and the native GDSC Z-score were preserved for provenance. Concordance of the harmonized metric across cohorts was assessed on the shared cell lines and drugs by the per-drug Spearman correlation between GDSC and CTRP **sens_z**.

2.4 Expression merge and multi-omic layers

The modeling universe was defined as every cell line with both a CCLE expression profile and at least one harmonized drug measurement. Expression was stored as $\log_{10}(\text{TPM})$ with Ensembl version suffixes stripped. TCGA Disease labels were assigned per cell line from the CCLE. Eight companion multi-omic layers were cleaned and aligned to the same cell-line universe by DepMap identifier, with feature counts and coverage as follows: mutations restricted to a 71-gene cancer-driver panel (binary, 802 of 810 lines covered), 517 recurrent gene fusions (binary, 810 lines), promoter methylation at 9 informative transcription-start-site windows (726 lines), 225 metabolites (799 lines), 214 RPPA antibodies (781 lines), 734 mature microRNAs (805 lines), 42 global chromatin marks (781 lines), and a 4-feature ABSOLUTE ploidy and whole-genome-doubling summary (786 lines). Binary genetic layers stored profiled-negative cells as explicit zeros, so that absence from a layer denotes missing data rather than a negative call; no imputation or standardization was applied at the storage stage. Cleaned layers were reproduced against a previously validated implementation of the same cleaning rules (agreement of 100% for binary and 99.9% for continuous layers); this is a consistency check rather than an external validation, and it was accompanied by spot checks against known biology (e.g., near-ubiquitous *KRAS* mutation in pancreatic lines and top-ranked *MGMT* promoter methylation in glioblastoma).

2.5 Predictive modeling and evaluation

All models predicted `sens_z`. The pooled design matrix comprised principal components of the per-panel expression matrix concatenated with a one-hot encoding of drug identity; cell-line identifier served as the grouping variable and drug identifier as the residualization/label variable. Model families were implemented in scikit-learn [21]: an elastic-net regressor [22], a drug-mean regressor (predicting each drug’s mean sensitivity from the one-hot block while ignoring molecular features), and a training-mean floor. Optional multi-omic blocks were appended after the drug one-hot under a base-impute-and-flag scheme: an unmeasured binary feature was set to zero, an unmeasured continuous feature was set to the mean of that column in the training fold, and a per-layer indicator recorded whether the cell line had been profiled on that layer at all, so that missingness itself remains available to the model rather than being silently imputed away. The join is response-agnostic and therefore leakage-free.

Evaluation used **grouped, leave-cell-line-out nested cross-validation**: a given cell line never appeared in both training and test folds, so the reported performance reflects generalization to unseen lines. A random row-wise split would leak, because each cell line contributes one row per drug it was screened against: a model could learn a line’s general level of chemosensitivity from its training rows and reapply it to that same line’s held-out rows, scoring well without having learned anything that transfers to a new line. Hyperparameters were selected in inner folds; performance was scored by held-out Spearman correlation between predicted and observed sensitivity, and repeated cross-validation shuffles were used to form confidence intervals. Every configuration was scored on identical splits against the drug-mean baseline.

To separate the two components of the prediction problem, the mechanism-specific arm used **in-fold residualization**: within each outer training fold, per-drug means were computed on training rows only and subtracted from both the fit target and the (train-mean-referenced) test target, so the model was asked to predict the personalization signal that remains after drug main effects are removed. A held-out residual Spearman correlation whose confidence interval excludes zero indicates genuine mechanism signal; a value near zero indicates that molecular features only recover drug main effects. Because the per-drug means are computed strictly in-fold, this procedure does

not leak.

Removing the per-drug mean is, however, nearly a no-op pan-cancer, because `sens_z` is already standardized per drug; the operative confound once tissues are pooled is tissue of origin, which expression encodes as a near-perfect lineage readout. To separate tissue-of-origin structure from any remaining within-line signal, the mechanism-specific arm was therefore additionally evaluated under **disease-by-drug residualization**. The within-tissue permutation null used to assess it reassigns expression between lines of the same tissue, so it controls lineage specifically and not every line-level property that could covary with sensitivity. Within each outer training fold, the mean sensitivity of every (disease, drug) cell was computed on training rows only and subtracted from both the fit target and the train-referenced test target, using a graded fallback for cells whose lines all fell in the test split, in which the (disease, drug) mean was replaced by that drug’s overall mean and, failing that, by the global mean; the analysis was restricted to (disease, drug) cells with at least five distinct cell lines (retaining 96.9% of rows) so that each cell mean is estimable, and the identical row set was used for every arm so that any change in performance reflects tissue removal rather than a change of samples. Whatever held-out correlation survives this subtraction estimates personalization beyond *both* drug identity and tissue of origin. Because a repeated-cross-validation interval measures split-stability rather than sampling noise (Section 3.5), significance for this arm was assessed against a **within-tissue permutation null**: within each tissue, the mapping from cell line to its expression profile was permuted (100 permutations), holding tissue and drug structure and the (disease, drug) cell means fixed while breaking only the correspondence between a line and its own expression; the permutation p -value and z -score are reported. As a no-feature reference, a **tissue baseline** predicted the drug-residualized sensitivity from the in-fold (disease, drug) cell mean alone, quantifying the held-out correlation obtainable from lineage identity alone, without any molecular features.

Three cohort settings (GDSC-only, CTRP-only, and pooled GDSC+CTRP) were compared to separate two effects that pooling the two screens could have: narrowing the confidence interval around an estimate, and raising the estimate itself above the drug-mean baseline. Models were evaluated both **per disease** (each tissue analysed separately, restricted to diseases with at least 20 lines) and **pan-cancer** (all tissues pooled into a single panel), the latter to test whether increasing effective sample size lifts the mechanism arm.

2.6 Pathway features and per-drug signatures

To test whether pathway biology captures mechanism signal that raw expression misses, $\log_{1p}(\text{TPM})$ expression was mapped from Ensembl identifiers to gene symbols (collapsing duplicates by maximum) and scored using single-sample gene-set enrichment analysis (ssGSEA) as implemented in GSVA 1.50.5 [23, 24, 25] (R 4.3.3, Bioconductor 3.18) against Reactome gene sets [26] retrieved through the msigdb 26.1.0 interface to the Molecular Signatures Database [27]. This yielded a label-blind pathway-activity matrix (1,808 Reactome pathways $\times$ cell lines) that was substituted for, or reduced to principal components alongside, the gene-expression features in the modeling harness (the pathway-activity analysis).

A complementary per-drug signature analysis fitted, for one (disease, drug) panel at a time, a supervised gene signature: Spearman correlations between each gene and sensitivity were computed on the training fold only, the top-ranked genes were combined into a signature score, and held-out cells were projected; the signature was refit within every fold inside a single pipeline, so selection is leakage-free. Because supervised top- k selection over thousands of genes can appear predictive even on shuffled labels, each per-drug signature was assessed against a **permutation null**: the sensitivity

vector was shuffled (100 shuffles), the entire in-fold procedure re-run, and an empirical permutation p -value computed. Per-drug claims are reported using this permutation p -value rather than the repeated-cross-validation confidence interval. With 100 shuffles the smallest attainable empirical p -value is $1/101 = 0.0099$, so a reported $p = 0.010$ denotes a signature that no permutation matched rather than a precisely estimated tail probability. To test the recovery of established pharmacology prospectively rather than on pairs chosen after the fact, a **mechanism-defined reference set** was fixed before inspecting any result, as every screened combination of seven target classes (BRAF, MEK, EGFR/HER2, BCR-ABL/SRC, BCL2, IGF1R, MDM2/p53) with the tissues in which that class’s target is an established dependency; the validation rate within this set was compared with the background rate across all panels by a one-sided binomial test. This analysis was run at full scale over **all 7,836 eligible (disease, drug) panels** (≥ 20 lines and minimum sensitivity variance per pair, across 20 diseases); the pathway-activity ablation was likewise carried to a permutation-controlled sweep over the 15 disease panels that support the Reactome layer. Both are parallel problems and were executed on the distributed batch fan-out described in Section 2.7.

Two further reference distributions were computed so that claims of chance-level performance and of non-random recurrence rest on explicit nulls rather than on intuition. First, the drug-ranking metric (normalized discounted cumulative gain over the three top-ranked drugs, computed with graded relevance and averaged over cell lines) has a non-zero expectation under a random ranking, so a **random-ranking null** was formed by permuting each cell line’s own measured sensitivities, which breaks the correspondence between a drug and its response while preserving that line’s sensitivity distribution exactly; the null was evaluated over all 836 cell lines in the harmonized response store with at least ten measured drugs (a no-feature null needs no expression, so it is not restricted to the 810-line modeling universe of Section 2.4), 100 permutations each. Second, because a compound tested in many tissues has many opportunities to be called significant, a **recurrence null** was built for the cross-tissue analysis: each tissue’s hits were redrawn without replacement from the drugs actually tested in that tissue, holding both the per-tissue hit count and each drug’s set of tested tissues fixed, and the number of drugs recurring in at least k tissues was recounted (2,000 permutations). A second version additionally matched the draws on panel size, sampling within quartiles of the number of cell lines per panel, so that recurrence driven by statistical power rather than shared biology is absorbed into the null; both versions are reported.

Finally, an interpretation-only enrichment atlas was built by ranking genes by their correlation with sensitivity for each of 2,132 (disease, drug) panels and running pre-ranked fast gene-set enrichment analysis (fgsea 1.28.0) [28] against Reactome [26]. Because these rankings are computed on all cells, the atlas is used only to interpret and prioritize hits, never as a model feature (which would leak). As a complementary, independently run analysis, per-disease differential-expression sensitivity signatures were also generated on raw counts with DESeq2 [29] followed by pre-ranked Reactome enrichment.

2.7 Computational implementation: distributed batch fan-out

Modeling was implemented in Python (NumPy 2.4.4, SciPy 1.17.1, pandas 3.0.2, scikit-learn 1.8.0, pyarrow 18.1.0, joblib 1.5.3) [21, 30, 31]; pathway feature generation used R (GSVA, fgsea, msigdb). Iterative development ran on a single multi-core cloud instance, but the full permutation sweeps (the full 7,836-panel per-drug signature analysis and the 15-panel pathway-activity ablation, each with a 100-shuffle permutation null) are embarrassingly parallel and were run as a stateless map/reduce on a transient Kubernetes cluster.

An Amazon Elastic Kubernetes Service (EKS) cluster (Kubernetes 1.34) was created per session

and was torn down after a sweep ran. Its worker fleet was a managed spot node group of eight 16-vCPU arm64 (AWS Graviton) instances drawn from several instance families to widen the spot pool (128 vCPU total), created at zero capacity and scaled up only for the fan-out. A single container image (built natively for arm64, Python 3.11) carried the analysis code; Argo Workflows v3.6.2 [32] enumerated the independent work units and fanned out **one Kubernetes pod per shard** via a workflow parameter. Amazon S3 was the sole data plane: each pod synced its inputs down from S3, ran exactly one shard, and synced a per-shard checkpoint back up, so a pod that was interrupted (spot reclamation) or failed (e.g., out-of-memory) simply re-ran its shard, and a shard whose checkpoint already existed short-circuited, making the whole job idempotent and safe to resubmit. A final single-process reduce pass globbed the checkpoints into the combined result table. Pods requested CPU and memory equal to their limits (Guaranteed quality of service) for predictable bin-packing, and the numerical-library thread count was pinned to one per process (OpenBLAS/MKL/OMP) so that all parallelism lived at the joblib worker level with the worker count set equal to each pod’s CPU request; without this pinning a packed node oversubscribes its cores and a single shard’s permutation loop slows by more than an order of magnitude.

The two workloads differed only in how work was binned and in the per-pod resource envelope. The pathway ablation ran one {disease} panel per pod (15 pods, 7 vCPU and 8 GiB each), so all shards ran concurrently and wall-clock was set by the slowest panel (~1.5–2 h). The per-drug signature sweep enumerated 7,836 {disease, drug} pairs and balanced them into **322 shards of ≤ 25 drugs within a single disease** (5 vCPU and 8–16 GiB each), so each pod staged a disease’s expression store once and looped its drugs; with 24 pods resident at a time the 322 shards drained in successive waves. AWS coordinates, exact pod specifications, and submission commands are recorded with the released code. All derived stores are keyed on Cancer Dependency Map (DepMap) identifiers, every stage writes checkpointed intermediate outputs, and code and processed data are archived for reuse (see Data and Code Availability).

2.8 Transfer-robust mechanism matching (Semblance 1.0)

To support recommendation for molecular profiles generated outside the training cohort (where raw gene-expression values are not directly comparable across sequencing pipelines and batches), the mechanism-specific arm is designed to be queried through a rank-based semantic-matching engine, *Semblance*, rather than through raw-gene signatures. The interpretation-only enrichment atlas (Section 2.6), 2,132 per-panel Reactome enrichment tables in standard gene-set-enrichment output format, serves as a reference library of drug-mechanism signatures; 1,824 of these tables form the served library, the remainder being dropped because the screened compound is an uninterpretable screening identifier or a non-therapeutic control agent (306 panels) or because no sensitivity-associated pathway passes the thresholds below (2 panels). *Semblance* represents each enrichment table as a centroid of biomedical text embeddings (BioLORD-2023 [33]) of its pathway names, weighted by absolute normalized enrichment score and split by enrichment direction, and scores a query enrichment table against the reference library by cosine similarity of these centroids (with per-pathway enrichment-score rank correlation and leading-edge gene Jaccard overlap reported as baselines). Because the representation is built from pathway *meaning* rather than gene magnitudes, it is intended to be robust to the cross-platform scale and batch effects that degrade raw-expression transfer, a design property that is not tested here, and a user’s arbitrary gene-set collection still aligns with the Reactome atlas even across differing ontologies. A high match between a query’s up-regulated program and a drug’s atlas signature (which is oriented so that positive enrichment marks programs elevated in *sensitive* cells) reads as “this profile resembles cells sensitive to that drug.” In this manuscript *Semblance* is used only as the mechanism-matching

layer of the accompanying recommendation tool and to motivate the transfer-robustness argument in the Discussion; the drug-sensitivity benchmark itself does not depend on it. *Semblance 1.0* is released as a hosted tool for pairwise semantic comparison of two gene-set-enrichment results that remains available at <https://huggingface.co/spaces/yueyvettehao/Semblance>.

2.9 An open recommendation application (*Semblance 2.0*)

The harmonized resource, the confidence-tiered evaluation, and the semantic-matching engine are delivered together as an open web application, *Semblance 2.0*, whose architecture follows the two confidence tiers. The application is split into a dependency-free static frontend and a single stateless compute service. The **Cell Line Lookup** function is fully static: for each of the 810 profiled cell lines a compact artifact was precomputed from the merged store, holding the within-cancer-type broad-spectrum ranking and, where they exist, the permutation-validated mechanism-specific signatures ($p \leq 0.01$), so that this tier is served directly with no server. The **Signature Match** function is backed by a small CPU-only service (a FastAPI application) that holds the 1,824-signature enrichment atlas and the *Semblance* engine of Section 2.8; a submitted enrichment table is parsed, embedded, and matched against the atlas at request time.

Retrieval against a library of this size required one departure from the pairwise engine of Section 2.8, where a single similarity is reported between two tables and no ranking is involved. Ranking a query against 1,824 candidate signatures cannot be done on raw cosine, because the embedding space is strongly anisotropic: the mean pairwise cosine among the atlas centroids is 0.92, an arbitrary pathway program scores ≈ 0.975 against any disease panel, and a minority of atlas entries are the nearest neighbour of almost any query. Raw-cosine ranking is correspondingly insensitive to the input, with two independent arbitrary programs returning 75% of the same top-20 drugs. Matches are therefore ranked on a hubness-corrected, null-calibrated statistic: cross-domain similarity local scaling [34] subtracts each atlas entry’s mean similarity to arbitrary programs, the residual is expressed as a z-score against a null built from real enrichment tables held out panel-wise, and that value is standardized within the panel being ranked so that it is comparable across queries of differing internal coherence. On a held-out retrieval check, submitting a drug’s own enrichment table and asking where that drug is returned, this recovers the correct drug at rank 1 for 95% of 74 signatures across the melanoma, breast and colorectal panels (100% within the top five), against 68% under raw cosine, while the top-20 overlap between independent arbitrary programs falls to 27% against a 26% chance floor. A per-query confidence statistic was also examined and is deliberately not reported: candidate calibrations failed to separate a drug’s own signature from an arbitrary program in held-out controls, so the application ranks candidates without asserting per-query significance, and the raw cosine is retained in the output for transparency only.

The atlas centroids and the precomputed BioLORD embeddings of the Reactome vocabulary are bundled with the service, so that a query whose pathway names are already in the atlas is scored without loading the ≈ 420 MB embedding model at all; pathway names not in the atlas fall back to loading it lazily. The static frontend is hosted on GitHub Pages at <https://yueyvettehao.github.io/Semblance2.0/> and the backend matching service as a container on Google Cloud Run. *Semblance 2.0* supersedes the earlier *Semblance 1.0* and reuses *Semblance*’s deterministic enrichment-comparison core as the matching layer of its Signature Match function. Both the application and the underlying engine are released as open source (see Code Availability).

3. Results

3.1 A structure-harmonized GDSC∪CTRP–CCLE resource

Harmonizing the two screens on canonical identifiers produced a single long-format store, the first stage of the pipeline summarized in *Figure 1*. GDSC2 contributed 234,757 measurements (967 lines $\times$ 285 compounds) and CTRP v2.2 contributed 253,042 measurements (645 lines $\times$ 479 compounds). Their union spans **678 drugs across 1,171 cell lines (487,799 measurements)**; the structure-confirmed intersection, used for quality control, comprises 86 drugs across 441 lines (37,361 co-measured drug–cell-line pairs, 66,738 measurements across the two cohorts). Resolving drug identity on the InChIKey connectivity block rather than on name was consequential: it raised the shared-compound count from 71 to 86 and, in the reverse direction, prevented three same-name/different-structure pairs from being incorrectly pooled (*Figure S1a*; *Table 1*).

The harmonized sensitivity metric is concordant across the two independent screens in the expected, modest way. Across shared drugs, the per-drug Spearman correlation between GDSC and CTRP `sens_z` had a median of **0.385, with 95% of shared drugs correlating positively**, consistent with the known moderate concordance of the two studies [15] and indicating that the sign and scale of the pooled metric are correct (*Figure S1b*). As a concrete check, the most-sensitive lines for bortezomib were multiple myeloma lines, as expected for a proteasome inhibitor.

Joining the harmonized response to CCLE expression yielded the modeling universe: **810 cell lines** (613 present in GDSC, 631 in CTRP, 434 in both) profiled across **57,820 genes** and spanning **37 diseases**. The eight multi-omic layers were aligned to this same universe, with coverage varying by disease and by assay (*Figure S1d*). The central statistical constraint of the study is visible here: most per-disease panels contain only 20–70 cell lines against tens of thousands of features, a severe $p \gg n$ regime (*Figure S1c*).

Table 1. The harmonized GDSC∪CTRP drug-response resource

Cell lines and compounds contributed by each screen, their union, and the structure-confirmed intersection used for cross-cohort quality control. Drug identity is the InChIKey connectivity block, not the compound name.

Set	Cell lines	Drugs
GDSC2	967	285
CTRP v2.2	645	479
Union	1,171	678
Shared	441	86

3.2 Broad-spectrum sensitivity is real but is tissue-by-drug structure

Within each disease, a drug-mean model (which ranks drugs by their average within-disease sensitivity while ignoring the molecular profile entirely) was strongly predictive. Held-out Spearman correlations were positive almost everywhere and largest in hematologic lineages (acute lymphoblastic leukaemia 0.54, diffuse large B-cell lymphoma 0.51, acute myeloid leukaemia 0.42) and in several solid tumours (renal 0.26, small-cell lung 0.25, sarcoma 0.24, and head-and-neck, pancreatic, and melanoma each ≈ 0.22), flat only in gastric and one pooled lung grouping (*Figure 2a*; *Table 2*). In other words, a substantial, reliable ranking signal exists, but it is a statement about which drugs are generically potent in a given tissue, not about the individual tumour.

Pooling makes the same point, though it cannot on its own establish it. When all tissues are combined into a single pan-cancer panel, the drug-mean model **carries no usable ranking signal** (pooled Spearman -0.029 , 95% CI $[-0.036, -0.021]$; -0.081 $[-0.091, -0.072]$ for CTRP-only; $+0.001$ $[-0.001, +0.002]$ for GDSC-only). Two of the three settings sit slightly but reliably below zero rather than at it, an artifact of the per-drug standardization described next. Because `sens_z` is standardized per drug across the whole panel, each drug’s pooled mean is ≈ 0 , so ranking by it carries no information once the tissue axis is removed. The strong per-disease performance was therefore tissue-by-drug structure (“this lineage is generically sensitive to this drug”), and the broad-spectrum signal is inherently a within-tissue quantity that does not survive as a pan-cancer ranking (*Figure 2b*). The same pattern holds for a ranking-quality metric: normalized discounted cumulative gain over the three top-ranked drugs, which asks how sensitive a line actually is to the drugs the model places first, is high within most diseases (0.42 to 0.69) but falls to 0.41 when tissues are pooled, indistinguishable from the 0.416 expected when each line’s drugs are ranked at random (Methods, Section 2.6) (*Figure S2*).

3.3 Expression personalization is minimal and largely recovered tissue of origin

The mechanism-specific arm (expression principal components predicting drug-mean-residualized sensitivity) tested whether molecular features personalize beyond drug main effects. **Per disease, this signal sat at zero.** In the pooled-cohort setting, only **1 of 20 diseases** (sarcoma, $+0.084$) had a residual Spearman confidence interval excluding zero from above; of the rest, most straddled zero and three (glioblastoma, bladder, and the pooled lung grouping) fell reliably below it, with a mean near 0 (*Table 2*). Fifty expression principal components did not crack personalization at these sample sizes, replicating, on harmonized data, the long-standing observation that general sensitivity levels and tissue structure dominate biomarker signal in these panels [16, 17].

Pooling the two screens estimated the same quantity more precisely without moving it away from zero. The mean 95% confidence-interval width of the mechanism arm narrowed from 0.114 (GDSC-only) to 0.102 (CTRP-only) to 0.100 (pooled, $\approx 12\%$ narrower), while the mean correlation stayed ≈ 0 . The larger combined sample therefore measures a near-absent personalization signal more sharply, rather than revealing one that either screen alone was too small to detect (*Figure 2c*).

Placing the two arms side by side across the per-disease and pan-cancer settings shows them trading places, the flip highlighted in *Figure 2b*: the broad-spectrum arm, strongly positive within every tissue (Section 3.2), falls to roughly zero when tissues are pooled, while the mechanism arm, flat within tissues, rises to $+0.10$. This reversal is an artifact of pooling rather than a genuine change in where predictive signal lives. The broad-spectrum arm collapses only because per-drug standardization forces each drug’s pooled mean to near zero, so within-tissue potency has nothing left to rank once the tissue axis is gone; the mechanism arm’s rise, examined next, proves largely to be recovered tissue of origin rather than new personalization. Pooling all tissues therefore buys effective sample size, but what it reveals cannot be interpreted without first separating lineage from personalization. Pan-cancer, the mechanism arm became clearly non-zero for the first time (pooled Spearman $+0.100$; CTRP-only $+0.128$; GDSC-only $+0.100$), with every confidence-interval lower bound above zero and an interval roughly $33\times$ tighter than per-disease (0.003 vs 0.100) on $\approx 398,000$ rows. However, this lift is **not yet shown to be personalization**. Because expression encodes lineage and lineage predicts sensitivity, a model that learns only that melanoma-like expression implies sensitivity to a particular set of drugs scores exactly this way, recovering through expression the same tissue-by-drug structure that the per-disease drug-mean model exploited directly. The pan-cancer $+0.10$ is therefore a mixture of true within-line personalization and recovered tissue-of-

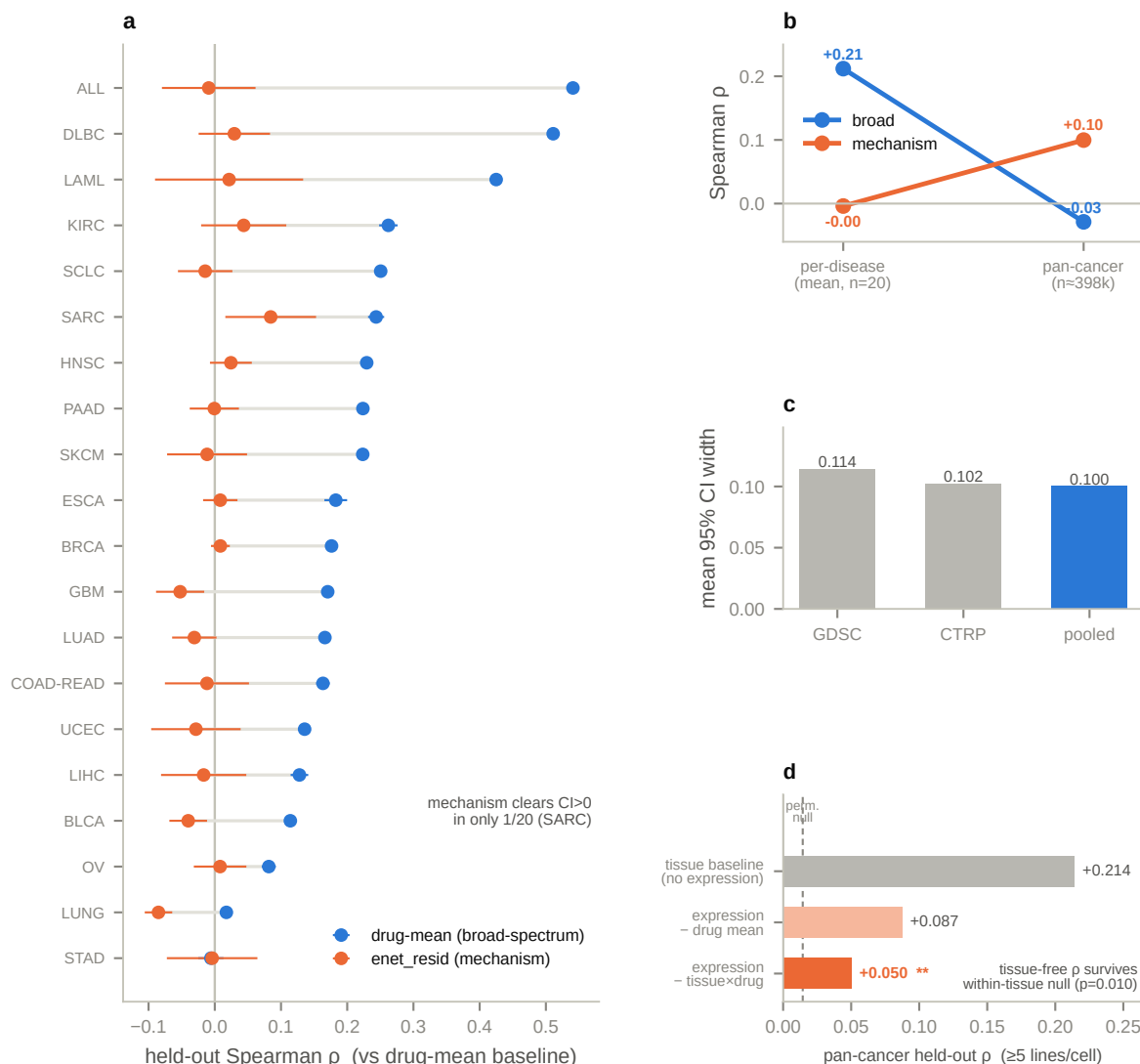


Figure 2. Broad-spectrum versus mechanism, the pan-cancer flip, and the tissue-free test. (a) Per-disease decomposition: the drug-mean (broad-spectrum) arm is positive in almost every tissue, flat only in gastric and one pooled lung grouping, while the expression mechanism arm clusters at zero (clearing CI>0 from above in only 1/20 diseases, sarcoma). (b) Pooling all tissues flips the two arms: broad-spectrum falls to ≈ 0 (-0.029 , CI $[-0.036, -0.021]$) while the mechanism arm rises to $+0.10$. (c) Pooling the two screens narrows the interval without raising the estimate above the baseline. (d) Disease-by-drug residualization removes tissue of origin from the target before the model is fit: on the ≥ 5 -line-per-cell row set the drug-residualized arm ($+0.087$) falls to $+0.050$ once tissue is removed, which survives a within-tissue permutation null ($p = 0.0099$, the floor of a 100-shuffle test; $z = 3.4$). The two expression bars are correlations against different, nested targets, so the gap between them is a drop in held-out skill and not a share of one signal. A no-feature tissue baseline ($+0.214$) outscores the expression arm on the drug-residualized target.

origin signal, and separating the two requires removing tissue from the target before the model sees it.

The pan-cancer analysis was therefore repeated under **disease-by-drug residualization** (Methods, Section 2.5), which removes tissue of origin from the target before the expression model ever sees it: within each training fold, the mean sensitivity of every (disease, drug) cell was subtracted from the target, and the analysis was restricted to (disease, drug) cells with at least five lines, identically across arms. On this row set the drug-residualized arm scores $\rho = +0.087$; after tissue is additionally removed, the pooled expression arm retains a held-out Spearman $\rho = +0.050$, which is significant against a within-tissue permutation null that reassigns each line’s expression to another line of the same tissue (permutation $p = 0.0099$, $z = 3.4$; no permutation exceeded the observed value) (Figure 2d). Removing tissue therefore costs roughly two-fifths of the held-out correlation (+0.087 to +0.050). That comparison is between two nested targets rather than a variance decomposition, so it should be read as the drop in held-out skill attributable to lineage, not as a partition of the signal into fixed shares. What survives is small, non-zero and permutation-validated, but the null it survives reassigns expression only within a tissue: it therefore excludes lineage, and does not exclude other line-level properties that vary within a tissue and track chemosensitivity, proliferation rate being the obvious one. The residual is best described as within-tissue, line-specific signal of unresolved origin rather than as established molecular personalization. The single-cohort runs agree in direction but reach significance only when the two screens are pooled (GDSC-only $\rho = +0.022$, $p = 0.08$; CTRP-only $\rho = +0.025$, $p = 0.07$), consistent with personalization being a small effect that needs the pooled effective sample size to detect.

Two further readings sharpen the picture. First, a no-feature **tissue baseline**, the in-fold (disease, drug) cell mean alone using no molecular features, scores $\rho = +0.214$ against the drug-residualized target, more than double the +0.087 of the expression arm; lineage identity therefore predicts the drug-residualized target *better* than fifty expression principal components do, so the expression model does not even fully recover tissue, let alone add personalization beyond it. Second, the ≥ 5 -line filter lowers the drug-residualized arm from the +0.100 reported above (computed on all rows) to +0.087 on the restricted set, while GDSC-only, least affected by the filter, reproduces +0.102, confirming that the change is the filter, not a shift in the underlying drug-only quantity. Taken together, the pan-cancer mechanism signal is real but modest: most of the apparent pan-cancer “personalization” is recovered tissue structure, and the personalization the study establishes on firmest footing remains the *per-drug*, permutation-validated signal of Section 3.5, shown within tissues against a proper null.

Consistent with the weakness of the personalization signal, naively adding multi-omic features hurt rather than helped. Appending 597 raw mutation, fusion, and methylation features to the expression principal components produced a mean-negative change in the mechanism arm (worst cases: oesophageal -0.095 , acute myeloid leukaemia -0.090 , gastric -0.060): at small n , high-dimensional genetic features dilute the estimator faster than L1 regularization can recover, arguing for feature-selected or pathway-prior genetics rather than a raw dump.

Table 2. Per-disease decomposition: broad-spectrum versus mechanism-specific

Held-out Spearman correlation (pooled GDSC+CTRP) for the drug-mean baseline, which ignores molecular features, and for the expression mechanism arm, which predicts drug-mean-residualized sensitivity. Brackets give the 95% repeated-cross-validation interval. The mechanism arm clears zero from above in one disease only (sarcoma); in three others (glioblastoma, bladder, pooled lung) the interval lies wholly below zero, and elsewhere it straddles zero.

Disease	Cell lines	Broad-spectrum ρ [95% CI]	Mechanism ρ [95% CI]
Acute lymphoblastic leukaemia (ALL)	23	+0.541 [+0.534, +0.548]	-0.009 [-0.080, +0.062]
Diffuse large B-cell lymphoma (DLBC)	26	+0.511 [+0.505, +0.517]	+0.030 [-0.024, +0.083]
Acute myeloid leukaemia (LAML)	21	+0.425 [+0.416, +0.434]	+0.022 [-0.090, +0.133]
Renal (KIRC)	21	+0.262 [+0.248, +0.276]	+0.044 [-0.021, +0.108]
Small-cell lung (SCLC)	42	+0.251 [+0.241, +0.260]	-0.015 [-0.056, +0.027]
Sarcoma (SARC)	30	+0.244 [+0.231, +0.256]	+0.084 [+0.016, +0.153]
Head and neck (HNSC)	33	+0.229 [+0.224, +0.235]	+0.024 [-0.007, +0.056]
Pancreatic (PAAD)	37	+0.224 [+0.218, +0.229]	-0.001 [-0.038, +0.036]
Melanoma (SKCM)	53	+0.223 [+0.219, +0.227]	-0.012 [-0.072, +0.049]
Oesophageal (ESCA)	25	+0.183 [+0.166, +0.200]	+0.008 [-0.018, +0.034]
Breast (BRCA)	49	+0.176 [+0.168, +0.185]	+0.008 [-0.006, +0.023]
Glioblastoma (GBM)	33	+0.171 [+0.165, +0.176]	-0.052 [-0.089, -0.016]
Lung adenocarcinoma (LUAD)	69	+0.166 [+0.163, +0.170]	-0.031 [-0.064, +0.003]
Colorectal (COAD/READ)	49	+0.163 [+0.153, +0.174]	-0.012 [-0.075, +0.052]

Disease	Cell lines	Broad-spectrum ρ [95% CI]	Mechanism ρ [95% CI]
Endometrial (UCEC)	27	+0.136 [+0.127, +0.144]	-0.028 [-0.096, +0.039]
Liver (LIHC)	24	+0.128 [+0.115, +0.141]	-0.017 [-0.081, +0.048]
Bladder (BLCA)	25	+0.114 [+0.108, +0.120]	-0.040 [-0.069, -0.012]
Ovarian (OV)	38	+0.082 [+0.071, +0.092]	+0.008 [-0.032, +0.048]
Lung (pooled) (SITE_LUNG)	20	+0.017 [+0.008, +0.027]	-0.085 [-0.106, -0.064]
Gastric (STAD)	32	-0.006 [-0.025, +0.013]	-0.004 [-0.072, +0.064]

3.4 Pathway re-representation does not unlock mechanism signal

Re-representing expression as Reactome pathway activity did not rescue the mechanism arm. This was tested at full permutation-controlled scale: a distributed fan-out (Methods, Section 2.7) ran a grouped nested cross-validation *and* a 100-shuffle permutation null for every one of the **15 disease panels** that carry enough cell lines for the 1,808-feature Reactome ssGSEA layer, comparing three pathway encodings against the gene-principal-component mechanism arm.

The verdict is a wash by any practically meaningful standard. A principal-component representation of the ssGSEA matrix (matched in dimensionality to the gene-PC arm) cleared a confidence interval above zero in **0 of 15** diseases, no better than the gene-PC arm, which itself cleared in only 1 of 15 (sarcoma, +0.084). The raw ~1,800-feature pathway model fared slightly better against the permutation null: significant at $p < 0.05$ in **5 of 15** diseases (sarcoma $p = 0.010$, breast $p = 0.010$, glioblastoma $p = 0.040$, ovarian $p = 0.030$, lung adenocarcinoma $p = 0.030$). These p -values are not comparable in strength to the per-drug ones quoted later: the pathway arm’s permutation null is far narrower (standard deviation 0.006 to 0.010, against ≈ 0.2 for a per-drug gene signature), so an identical printed p reflects a much smaller absolute margin. Even so, the effect sizes are negligible in all but two panels: only sarcoma (+0.19) and breast (+0.08) exceed a held-out Spearman of 0.05, while the other three “significant” diseases sit at +0.01 to +0.02. A global pathway re-encoding of expression therefore does not, by itself, expose personalization that the raw features miss; where a permutation-clean pathway signal exists at all, it is confined to a couple of tissues and is too small to matter for recommendation (*Figure S3*).

3.5 Two-stage significance control isolates a high-confidence set of per-drug biomarkers

The permutation-controlled per-drug sweep was run across **every eligible (disease, drug) panel**: all 7,836 pairs spanning 20 diseases (≥ 20 cell lines and minimum sensitivity variance per pair), each fitted with an in-fold supervised gene signature under grouped cross-validation and assessed against a 100-shuffle permutation null. This is the parallel workload that the distributed fan-out (Methods, Section 2.7) was built for. Across the full set, the in-fold signature beat a per-drug principal-component elastic net in **6,476 of 7,836** panels (83%), showing at scale that drug-specific supervised gene selection outperforms global components for drug-specific biomarkers.

Because significance here depends on how the null is defined (*Figure 3a*), the candidate set was refined in two stages, each raising the confidence attached to what remains.

The first stage replaced a split-stability criterion with a calibrated null. The heuristic in common use, that the repeated-cross-validation confidence-interval lower bound exceeds zero, admits **3,284 of 7,836** panels (42%). That interval quantifies how stable a prediction is across resampled splits rather than the sampling variability of the statistic under the null, and supervised top- k selection over roughly 8,000 genes can score positively on shuffled labels, giving a wide true null (standard deviation ≈ 0.2 at $n \approx 50$; *Figure 3b*). Re-scoring every panel against a permutation null (shuffled sensitivity, empirical p -value) retains **971 of 7,836** panels (12%) at $p < 0.05$, a **3.4-fold** reduction. The retained set is the one on which every per-drug claim below rests.

The second stage calibrates what survives the first. Roughly 392 of the 971 are expected by chance at this threshold (false-discovery proportion ≈ 0.4), so a single panel is treated as a candidate rather than a finding, and confidence is built by requiring corroboration from a source the permutation test does not use: agreement with established pharmacology, recurrence of the same compound across independent tissues, and coherent grouping by mechanism, each examined below. Together the two stages define the rule adopted throughout, that per-drug biomarker claims rest on the permutation p -value and its independent corroboration rather than on a confidence-interval lower bound.

The surviving associations are mechanistically coherent, well-established biomarkers rather than statistical curiosities (*Figure 3c*, *Table 3*). They recover the canonical targeted-therapy pairings across every major class: the BRAF inhibitors PLX-4720 (melanoma Spearman +0.50, $p = 0.010$; colorectal +0.45; also pancreatic and small-cell lung) and dabrafenib (melanoma +0.44) [35]; the MEK inhibitors trametinib and selumetinib in melanoma (+0.57 and +0.64); EGFR/HER2 inhibitors lapatinib and erlotinib in breast (+0.50, +0.48); insulin-like growth factor 1 receptor (IGF1R) inhibitors in the expected tissues (BMS-754807 in sarcoma +0.53 and breast +0.35; linsitinib and NVP-ADW742 in sarcoma and ovarian); the MDM2/p53 antagonist nutlin-3 in sarcoma (+0.60), colorectal (+0.32, $p = 0.020$) and acute lymphoblastic leukaemia (+0.53), consistent with p53-pathway dependence [36]; and the BCL2 inhibitor ABT-199 (venetoclax) in acute lymphoblastic leukaemia (+0.62). Recurrent multi-tissue hits included BI-2536 (PLK1), the PARP inhibitors talazoparib/olaparib/niraparib (strongest in sarcoma), dasatinib (SRC/ABL), vorinostat (HDAC), and AZD7762/AZD8055 (CHK1/mTOR), all mechanism-carrying targeted agents. These recoveries are the first of the two corroborations, and because the pairs quoted above were selected for mention *after* they validated, the corroboration was also tested prospectively. A reference set was fixed from target biology alone, before consulting any result, as every screened combination of seven drug classes with the tissues in which that class’s target is an established dependency (BRAF, MEK, EGFR/HER2, BCR-ABL/SRC, BCL2, IGF1R, MDM2/p53; Methods, Section 2.6). Of the 53 such panels actually screened, **28 (53%) validate**, against a background rate of 12.4% across

all 7,836 panels, a 4.3-fold enrichment (binomial $p = 1.5\text{e-}12$) (*Table S1*). The enrichment is uneven in an interpretable way: MDM2/p53 (5 of 5), BRAF (5 of 6) and IGF1R (6 of 9) validate strongly, whereas BCR-ABL/SRC (0 of 3) fails outright and several MEK and EGFR panels miss narrowly ($p = 0.06$ to 0.11), consistent with a mutation-defined responder subset too small to detect within a single tissue panel. Known pharmacology is therefore recovered at a rate far above chance but not exhaustively, and mechanism-specific prediction is viable at panel scale. It is concentrated in a well-powered subset of drug–tissue pairs (12%, roughly an eighth) rather than available uniformly for every drug.

The second corroboration, cross-tissue recurrence, points to where the new biology sits. Only 75 of the 971 permutation-validated hits involve the textbook targeted agents above; the remaining 896 are non-canonical, and 242 of the 428 distinct non-canonical drugs validate in two or more independent tissues, 57 of them in four or more (*Figure 4b*). Recurrence is only informative above a threshold, and the threshold is measurable. Under a null that redraws each tissue’s hits from the drugs actually tested there, holding both the per-tissue hit count and each drug’s set of tested tissues fixed, recurrence in two tissues is uninformative: across the full validated set 259 drugs recur in two or more tissues against 285 to 291 expected, so a drug screened in a median of 13 tissues has ample opportunity to be called twice by chance alone. The null is exceeded from three tissues upward, and the margin widens with order: 144 drugs recur in three or more tissues against 123 to 129 expected, 68 in four or more against 39 to 46, and 27 in five or more against 9 to 14 (permutation $p \leq 0.007$ at three tissues and $p < 0.001$ at four and five, under both the uniform and the panel-size-matched null; Methods, Section 2.6). Recurrence therefore carries evidential weight only above two tissues, and the mechanism groups examined below sit in the four-and-above range where the excess over the null is largest. Grouping the strongest recurrent non-canonical hits by putative mechanism (*Figure 4a*) surfaces coherent, largely novel vulnerabilities: ferroptosis induction (the glutathione-peroxidase-4 [GPX4] inhibitors RSL3, ML162 and ML210), BH3-mimetic apoptotic priming (ABT-737, WEHI-539, TW-37, sabutoclax), nuclear-export blockade (the exportin-1 [XPO1] inhibitors leptomycin B and KPT185), and mitotic and replication-stress checkpoints (the WEE1 inhibitor MK-1775, the Aurora inhibitors alisertib and VX-680, and the PLK1 inhibitor GSK461364). The clearest single example is melanoma ferroptosis: three chemically independent GPX4 inhibitors each yield a strong, permutation-validated melanoma signature (held-out $\rho = 0.63\text{--}0.71$) (*Figure 4c*), recovering, with no prior knowledge of the mechanism, the established ferroptosis vulnerability of dedifferentiated melanoma [37, 38]. A caveat tempers the promise: a large fraction of the recurrent non-canonical hits are broadly cytotoxic agents (topoisomerase poisons, antimetabolites, DNA-damaging drugs) whose signatures more plausibly track a general chemosensitivity or proliferation axis than a drug-specific mechanism (*shaded group, Figure 4a*). The mechanism-targeted classes are therefore the interpretable novel candidates, and every such hit is offered as a hypothesis for targeted follow-up given the per-panel false-discovery rate.

The enrichment atlas makes the biology underlying any one hit legible (*Figure S4*). For PLX-4720 in melanoma the sensitivity-associated program is dominated by melanocytic and neural-crest lineage pathways (a reminder that even a validated within-tissue signature can be lineage-linked rather than target-specific), whereas for the IGF1R inhibitor linsitinib in sarcoma it is a coherent DNA-replication and cell-cycle program, consistent with growth-factor-driven proliferation in the sensitive lines.

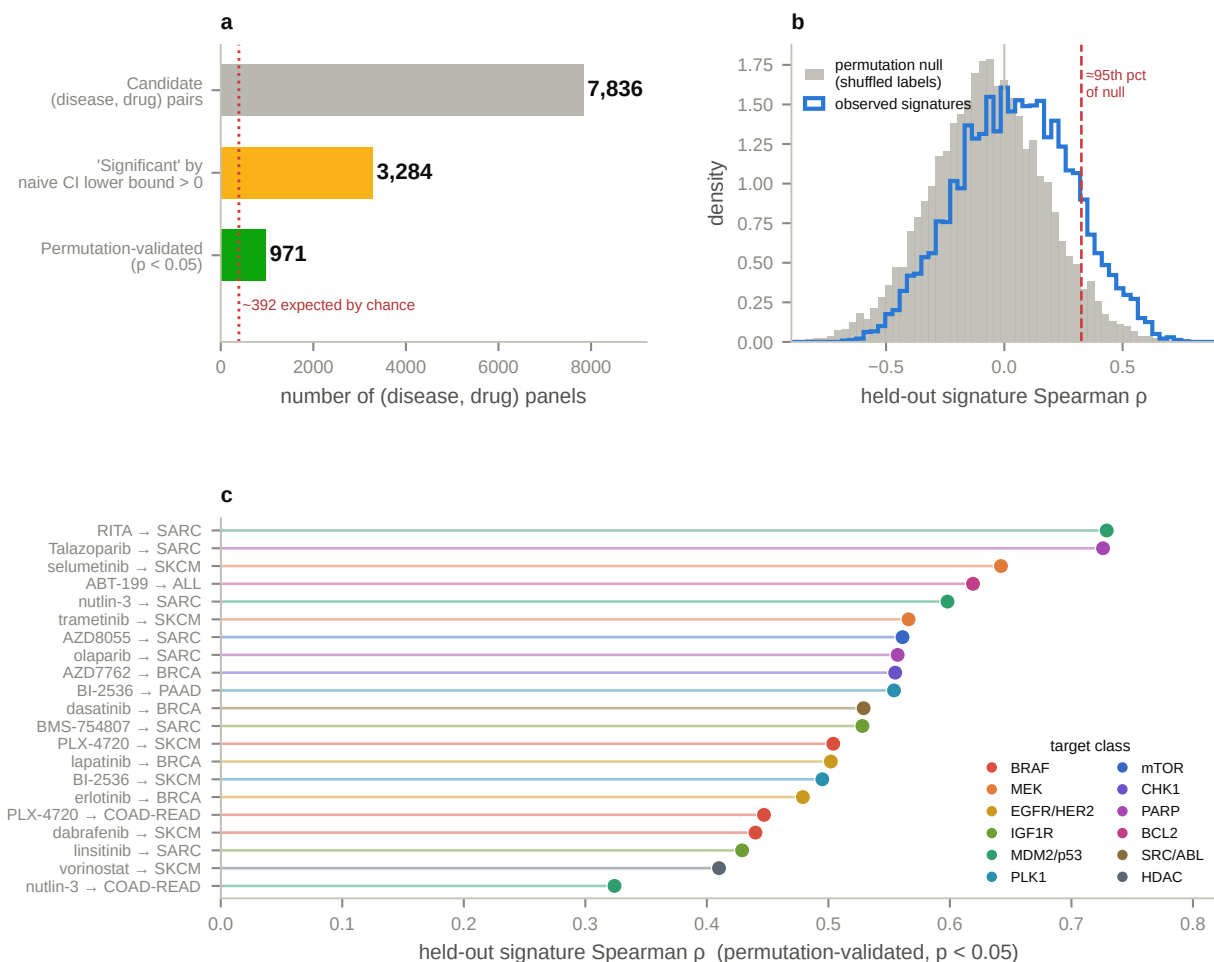


Figure 3. Per-drug signatures across the full permutation-controlled sweep (7,836 disease $\times$ drug panels). (a) Significance funnel: 3,284 panels pass the naive cross-validation interval and 971 survive a permutation null, a 3.4-fold difference; of those 971, ~392 are expected by chance at $p < 0.05$. (b) The permutation null is wide ($SD \approx 0.2$ at $n \approx 50$, 0.24 pooled across panel sizes), so a positive held-out ρ is not by itself evidence of signal. (c) The permutation-validated hits recover well-established biomarkers, coloured by target class; a reference set of drug–tissue pairs fixed in advance from target biology validates at 53% against a 12.4% background (4.3-fold, binomial $p = 1.5\text{e-}12$).

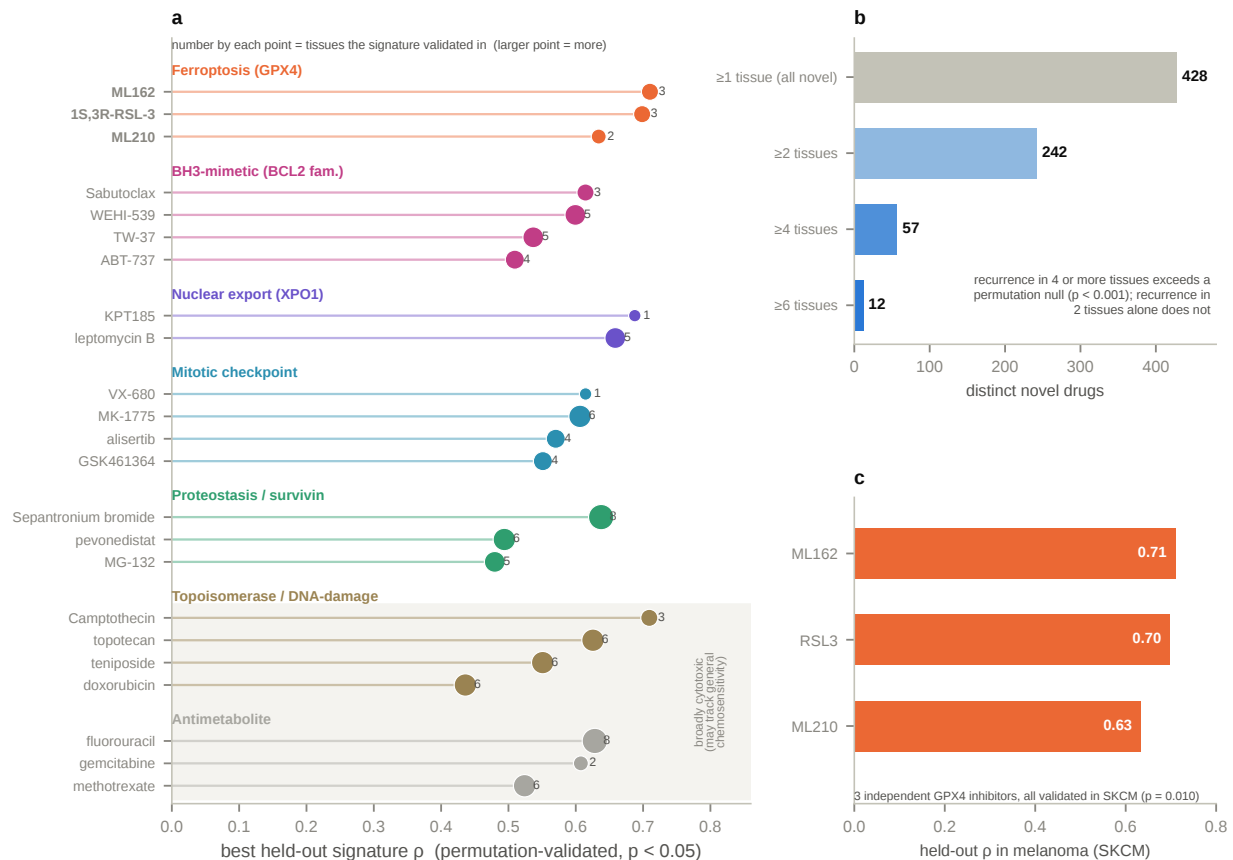


Figure 4. Novel, permutation-validated personalization signals beyond canonical biomarkers. (a) The strongest recurrent non-canonical validated hits, grouped by putative mechanism and ranked by best held-out signature ρ (point size = number of tissues the signature validated in); the mechanism-targeted classes (ferroptosis/GPX4, BH3-mimetic, XPO1 nuclear export, mitotic and replication-stress checkpoints) are the interpretable novel candidates, whereas the shaded broadly-cytotoxic classes may track a general chemosensitivity axis. (b) Recurrence spectrum: 242 of the 428 distinct non-canonical drugs validate in two or more independent tissues and 57 in four or more. Against a null that redraws each tissue's hits from the drugs tested there, computed on the full validated set, two-tissue recurrence is fully accounted for by chance (259 observed, 285–291 expected) whereas higher-order recurrence is not (three tissues: 144 observed, 123–129 expected, $p \leq 0.007$; four tissues: 68 observed, 39–46 expected, $p < 0.001$). (c) Melanoma ferroptosis: three chemically independent GPX4 inhibitors each yield a strong, permutation-validated melanoma signature (held-out $\rho = 0.63$ –0.71).

3.6 Summary of where signal lives

Taken together, the benchmark localizes predictive signal precisely. Broad-spectrum potency is strong and reliable but is a within-tissue property (Section 3.2). Expression-, omic-, and pathway-based personalization is essentially absent at current per-disease sample sizes; when it emerges pan-cancer it is small and mostly recovered tissue of origin: after disease-by-drug residualization only $\rho \approx 0.05$ survives (within-tissue permutation $p = 0.01$), and on the drug-residualized target a no-feature tissue baseline ($\rho = 0.214$) outscores the expression arm (+0.087) outright (Sections 3.3–3.4). Personalization is nonetheless present, concentrated rather than distributed: asked per drug and within tissue, and filtered by the two-stage protocol, 971 drug–tissue pairs (roughly an eighth of those tested) carry a permutation-validated, mechanism-specific biomarker (Section 3.5). The methodological corollary is quantitative and portable: across the full 7,836-panel sweep, the calibrated null retains a 3.4-fold smaller set than the conventional cross-validation interval admits, and that ratio is what separates a defensible discovery rate from an optimistic one.

Table 3. Representative permutation-validated per-drug biomarkers

Selected mechanistically coherent hits from the full per-drug signature sweep (held-out signature Spearman ρ ; permutation p ; n = cell lines). One representative panel per drug shown; many agents were significant in several tissues (see *Figure 3c*).

Drug	Putative target	Disease	n	ρ	perm p
PLX-4720	BRAF	Melanoma (SKCM)	53	+0.50	0.010
PLX-4720	BRAF	Colorectal (COAD/READ)	48	+0.45	0.010
dabrafenib	BRAF	Melanoma (SKCM)	45	+0.44	0.010
trametinib	MEK	Melanoma (SKCM)	46	+0.57	0.010
selumetinib	MEK	Melanoma (SKCM)	52	+0.64	0.010
lapatinib	EGFR/HER2	Breast (BRCA)	48	+0.50	0.010
erlotinib	EGFR	Breast (BRCA)	49	+0.48	0.010
BMS-754807	IGF1R	Sarcoma (SARC)	30	+0.53	0.010
linsitinib	IGF1R	Sarcoma (SARC)	30	+0.43	0.030
nutlin-3	MDM2–p53	Sarcoma (SARC)	30	+0.60	0.010
nutlin-3	MDM2–p53	Colorectal (COAD/READ)	48	+0.32	0.020
RITA	p53	Sarcoma (SARC)	28	+0.73	0.010
BI-2536	PLK1	Pancreatic (PAAD)	36	+0.55	0.010
AZD8055	mTOR	Sarcoma (SARC)	30	+0.56	0.010
AZD7762	CHK1	Breast (BRCA)	48	+0.56	0.010
talazoparib	PARP	Sarcoma (SARC)	24	+0.73	0.010
olaparib	PARP	Sarcoma (SARC)	30	+0.56	0.010
ABT-199 (venetoclax)	BCL2	ALL	23	+0.62	0.010
dasatinib	SRC/ABL	Breast (BRCA)	49	+0.53	0.010
vorinostat	HDAC	Melanoma (SKCM)	52	+0.41	0.010

Of 7,836 candidate (disease, drug) panels, 971 were permutation-validated at $p < 0.05$ (≈ 392 expected by chance). Full table in the released signature atlas.

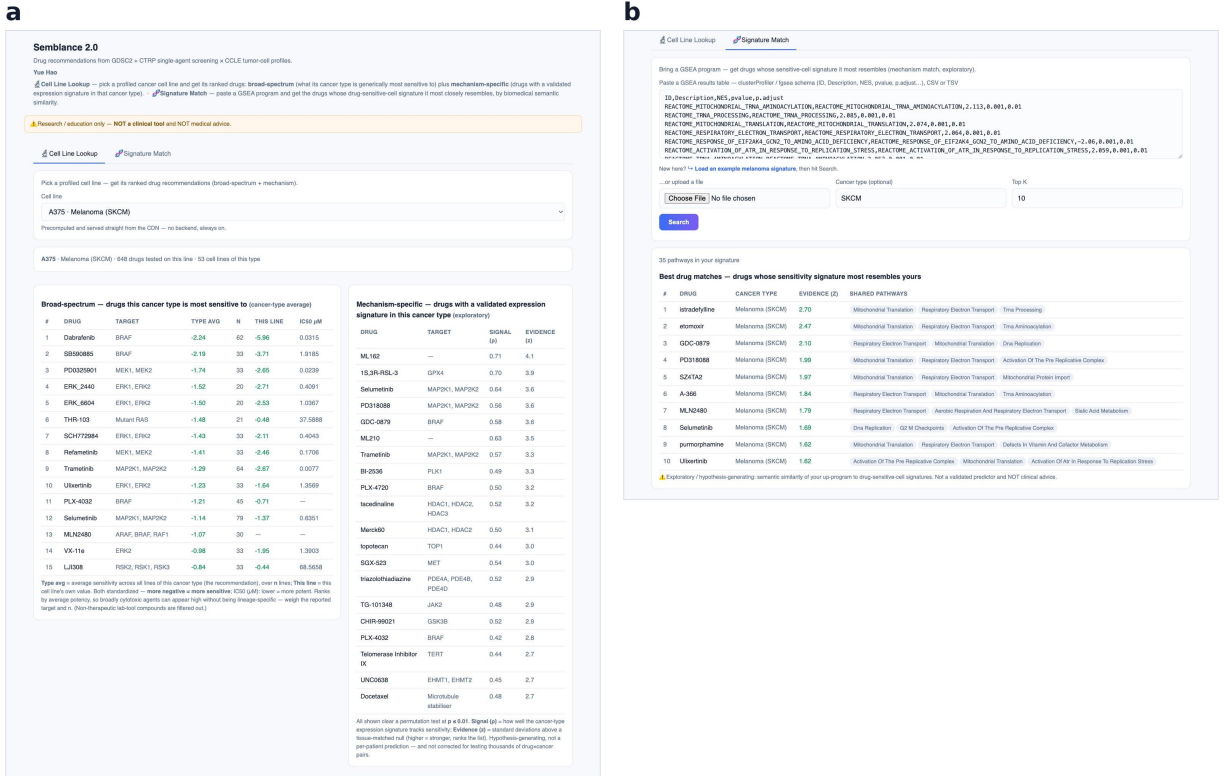


Figure 5. The Semblance 2.0 drug-recommendation application, a two-function open web tool built directly on the harmonized resource and the permutation-validated signatures. Both panels are captured from the live deployment at <https://yueyvetthao.github.io/Semblance2.0/>. (a) Cell Line Lookup: choosing a profiled cell line (here the BRAF-mutant melanoma line A375, 648 drugs tested across 53 lines of this cancer type) returns a confidence-tiered ranking — a high-confidence broad-spectrum arm (the drugs the cancer type is on average most sensitive to, here dominated by BRAF, MEK and ERK inhibitors: dabrafenib, SB590885, PD0325901 and two ERK inhibitors take the first five places) beside an exploratory mechanism-specific arm (drugs carrying a permutation-validated within-tissue expression signature at $p \leq 0.01$, ranked by evidence z). That second arm is led by the GPX4 ferroptosis inducers ML162 ($z = 4.1$) and 1S,3R-RSL-3 (3.9), with ML210 sixth (3.5), interleaved with MAPK-pathway inhibitors (selumetinib, PD318088, GDC-0879, trametinib) — recovering, from a single cell-line query, the melanoma-ferroptosis and BRAF/MEK biology established in Figures 3–4. (b) Signature Match: pasting an external gene-set-enrichment program (here the bundled example melanoma sensitivity program, 35 pathways) and optionally scoping it to a cancer type returns the drugs whose sensitive-cell atlas signature the query most resembles, with the shared pathways behind each match. Matches are ranked not on raw embedding cosine, which barely discriminates in this strongly anisotropic space, but on a hubness-corrected score referenced to a null of real held-out signatures and reported in standard deviations above it (Methods, Section 2.9). The top ten shown interleave MAPK-pathway agents (GDC-0879, PD318088, MLN2480, selumetinib, ulixertinib) with compounds whose association is not mechanistically obvious, and the leading shared pathways are largely generic proliferation and mitochondrial-translation programs — which is why this arm is presented as exploratory rather than as a recommendation. Both surfaces are labelled research/education-only and non-clinical. Cell Line Lookup is served as static precomputed artifacts; Signature Match calls an engine on Cloud Run over the 1,824-signature enrichment atlas.

3.7 An open, tiered recommendation tool: Semblance 2.0

The three findings above (a reliable within-tissue broad-spectrum signal, a permutation-validated per-drug mechanism signal for a minority of agents, and the discipline of scoring both against a proper null) are packaged into an open web application, *Semblance 2.0*, that turns the benchmark into a usable recommender while preserving its confidence tiering (Figure 5). The tool exposes two complementary functions. The first, **Cell Line Lookup** (Figure 5a), takes any of the 810 profiled cell lines and returns its drugs in the two tiers the benchmark supports: a high-confidence **broad-spectrum** ranking (the within-cancer-type mean sensitivity of Section 3.2) beside a deliberately separated, exploratory **mechanism-specific** list restricted to drugs with a permutation-validated within-tissue expression signature ($p \leq 0.01$; Section 3.5) and ranked by the null-referenced evidence z . For the BRAF-mutant melanoma line A375, for instance, the broad-spectrum tier is led by BRAF, MEK and ERK inhibitors, while the mechanism tier surfaces three chemically independent GPX4 ferroptosis inducers alongside MAPK-pathway agents, the same melanoma-ferroptosis and BRAF/MEK biology established in Figures 3–4, now reachable from a single cell-line query. The second function, **Signature Match** (Figure 5b), operationalizes the transfer-robust matching of Section 2.8: a user pastes an external gene-set-enrichment program and receives the drugs whose sensitive-cell atlas signature it most resembles by biomedical semantic similarity, with the shared pathways behind each match shown; scoping the query to a cancer type restricts the match to that tissue’s signatures. Because the match is computed on pathway meaning rather than raw gene magnitudes, it is intended to survive the cross-platform batch effects that degrade raw-expression transfer. Both surfaces carry an explicit research- and education-only, non-clinical disclaimer and preserve the sign convention (lower = more sensitive). Cell Line Lookup is served entirely as precomputed static artifacts, with no backend and therefore always available, whereas Signature Match calls a lightweight CPU-only engine over the 1,824-signature enrichment atlas; deployment and availability are detailed in Methods, Section 2.9 and the Code Availability statement.

4. Discussion

4.1 Where the predictive signal lives, and why

This study set out not to build a better drug-response predictor but to determine, under honest evaluation, where predictive signal genuinely resides once GDSC and CTRP are harmonized with CCLE and scored against a baseline that ignores molecular features. The answer becomes legible once sample size is treated as the organizing constraint rather than as a caveat. Per-disease panels hold 20 to 70 cell lines against tens of thousands of molecular features, and at that ratio the only effects that can be estimated reliably are those that are large and shared across many lines. Two such effects exist in these data, and the benchmark finds both: which drugs are generically potent, and which lineage a line belongs to. A drug-mean model is strongly predictive within every tissue yet carries no usable ranking signal when tissues are pooled, which identifies it as tissue-by-drug structure rather than personalization. Expression, multi-omic, and Reactome pathway features add essentially nothing beyond drug main effects at current per-disease sample sizes, clearing a confidence interval above zero in only one of twenty diseases across several feature representations (Figure 2a). That result is as much a statement about power as about biology, and it is what prior work would predict: general levels of drug sensitivity and tissue of origin dominate biomarker signal in preclinical panels [16, 17]. Personalization is by construction a smaller, line-specific quantity, and fifty expression components fitted on fifty cell lines cannot resolve it.

Pooling all tissues buys the effective sample size that per-disease panels lack, and it is the one

setting in which a molecular model becomes clearly predictive. What that model has learned is then the question, since expression encodes lineage and lineage predicts sensitivity. Residualizing sensitivity on the disease-by-drug mean removes tissue of origin from the target before the model is fit, and on that tissue-free target the pooled expression arm retains a held-out $\rho = +0.05$ against a within-tissue permutation null ($p = 0.01$), down from $+0.087$ on the same rows (*Figure 2d*). What survives that reduction is line-specific signal that is not lineage; because the null reassigns expression only within a tissue, however, it is not thereby shown to be molecular personalization rather than some other within-tissue line property such as proliferation rate. Two properties of this estimate set how it should be read: it reaches significance only when GDSC and CTRP are pooled, placing it near the limit of what $\sim 400,000$ measurements resolve, and a no-feature tissue baseline scores $\rho = +0.214$ against the drug-residualized target, where the expression arm reaches only $+0.087$, so lineage identity alone outpredicts fifty expression components. (The tissue baseline is not defined against the tissue-free target, since the in-fold disease-by-drug mean is precisely what that target removes.) Averaged over every drug and tissue at once, personalization is real but thin. That average is not the shape of the signal, however, and reading it as such would understate what the data contain. Asked one drug and one tissue at a time, where lineage is constant by construction and the estimate is not diluted across hundreds of compounds, the same question returns a much stronger answer: the two-stage protocol of Section 3.5 retains permutation-validated signatures for approximately 12% of the 7,836 drug-tissue panels, with held-out correlations of 0.32 to 0.73 across the representative validated panels of Table 3 (Section 3.5). The two findings are consistent with one reading, that personalization in these data is concentrated rather than distributed. Most drug-tissue pairs carry little line-specific signal, which is what pulls the pooled correlation toward zero, while a well-powered minority carries a great deal. The practical consequence is that personalization should be sought and reported per drug and within tissue, and that the pan-cancer correlation is best treated as a permutation-validated floor on the aggregate effect rather than as the quantity of interest.

4.2 A two-stage significance protocol for per-drug discovery

Beyond localizing signal, the per-drug analysis yields a transferable methodological contribution: an explicit, two-stage protocol for deciding which per-drug signatures to report. The permutation-controlled sweep was run across all 7,836 drug-tissue panels using the transient-cluster fan-out described in Methods, Section 2.7, so the resulting calibration is a property of the evaluation rather than of a favourable subset.

The first stage fixes what the significance test is measuring. A repeated-cross-validation interval quantifies the stability of predictions across resampled splits, not the sampling variability of the statistic under the null, and supervised feature selection over thousands of genes can score positively even on shuffled labels; judged by the common heuristic that such an interval excludes zero, 3,284 of 7,836 panels passed, whereas 971 passed a permutation null built by shuffling sensitivity within the panel (*Figure 3a*). The stakes are practical: this is the difference between reporting $\sim 42\%$ and $\sim 12\%$ of drug-tissue panels as carrying a validated biomarker, and only the second figure is defensible as a discovery rate.

The second stage attaches confidence to the retained set rather than to any single panel within it. Because approximately 392 of the 971 are expected by chance at $p < 0.05$, each panel is reported as a candidate and confidence is accrued from evidence the permutation test never sees: recovery of established drug-target pharmacology, recurrence of the same compound across independent tissues, and coherent grouping by mechanism (Section 4.3). Signatures that clear both stages

are the ones this study advances; those clearing only the first are not reported as biomarkers. Given how often cross-validation intervals serve as informal significance tests in this literature, the recommendation that follows is concrete: anchor performance to a feature-free baseline, gate discovery on a permutation null rather than on interval-excludes-zero, and promote surviving hits only on independent corroboration.

4.3 A minority of genuine, mechanism-specific biomarkers

The per-drug analysis also yields the study’s most valuable positive finding. After permutation control, candidate biomarkers survived for approximately 12% of drug–tissue panels (971 of 7,836) (*Figure 3c*, *Table 3*). Roughly 40% of that set is expected to be false, so it is a pool to be corroborated rather than a list of established findings, and the corroboration is quantitative: a reference set of drug–tissue pairs fixed in advance from target biology validates at 53% against a 12.4% background, a 4.3-fold enrichment ($p = 1.5\text{e-}12$). The surviving associations are correspondingly coherent rather than statistical noise: BRAF inhibitors (PLX-4720, dabrafenib) in melanoma and colorectal cancer [35], MEK inhibitors (trametinib, selumetinib) in melanoma, EGFR/HER2 inhibitors in breast, IGF1R inhibitors in sarcoma and breast, the MDM2/p53 antagonist nutlin-3 in sarcoma and colorectal cancer [36], and the BCL2 inhibitor venetoclax in acute lymphoblastic leukaemia, among many others spanning every major targeted-agent class. That supervised, in-fold per-drug signatures recover known pharmacology is reassuring evidence that the pipeline detects real biology where it exists, and it clarifies the shape of a defensible product: drug recommendation should be *tiered by confidence*. A high-confidence, broad-spectrum recommendation (rank a tissue’s drugs by mean sensitivity) is supportable today as a within-tissue statement; a lower-confidence, mechanism-specific recommendation is warranted only for the minority of drugs with a permutation-validated, expression-linked vulnerability, and should be labelled exploratory. Presenting a single, uniformly confident “personalized” ranking would misrepresent what these data support. That principle is implemented here rather than only argued: the harmonized resource, the permutation-validated signature atlas, and the two recommendation tiers are packaged into an open, confidence-tiered web application, *Semblance 2.0* (Section 2.9, *Figure 5*), which returns for any profiled cell line both a high-confidence broad-spectrum ranking and the exploratory, permutation-validated mechanism-specific list, and which matches a user’s own enrichment program to the drug-mechanism atlas by semantic similarity, turning “tier recommendations by confidence” into an interface rather than a recommendation.

More important than these canonical recoveries is what lies beyond them, because that is where genuinely new biology could come from. The permutation-validated set is dominated not by textbook pairings but by 896 non-canonical hits, 57 of whose distinct drugs recur across four or more independent tissues (*Figure 4b*); the null figures quoted next are computed across the full validated set, where the corresponding count is 68. Higher-order recurrence is the part that carries evidential weight. Against a null holding each tissue’s hit count and each drug’s tested tissues fixed, computed across the full validated set, recurrence in four or more tissues is observed at roughly half again the rate the null predicts (68 observed against 39 to 46 expected, permutation $p < 0.001$) and the excess persists at three tissues ($p \leq 0.007$), whereas two-tissue recurrence is entirely explained by chance, since a compound screened in a median of 13 tissues has many opportunities to be called twice. Reporting recurrence without that threshold would overstate the evidence, and the corroboration claimed here rests on the higher-order tail. Read by putative mechanism (*Figure 4a*), these surface coherent vulnerabilities that the canonical set misses: ferroptosis induction, BH3-mimetic apoptotic priming, exportin-1-mediated nuclear export, and mitotic and replication-stress checkpoints. The single cleanest example is melanoma ferroptosis, flagged independently by three chemically distinct

GPX4 inhibitors (*Figure 4c*), a de novo recovery of the established ferroptosis susceptibility of dedifferentiated melanoma [37, 38] that both validates the pipeline and illustrates its discovery value. The necessary caution is that a substantial fraction of the recurrent non-canonical hits are broadly cytotoxic agents whose signatures more plausibly encode a general chemosensitivity or proliferation axis than a drug-specific mechanism, so the mechanism-targeted classes are the credible candidates and each remains a hypothesis to be confirmed at the stated false-discovery rate. With that caveat, these candidates, not the small pan-cancer expression correlation of Section 4.1, are the study’s most actionable personalization signals, and they define a concrete experimental follow-up, such as testing GPX4 dependence in molecularly defined melanoma subsets.

4.4 Limitations

The principal limitations follow from the same analysis. First, the pan-cancer personalization signal, though now disentangled from tissue (Section 4.1), is small and fragile: it is significant only in the pooled cohort, its effect size ($\rho \approx 0.05$) is well under a quarter of the tissue baseline’s, and it is measured at the resolution of the TCGA tissue label, so finer lineage structure within a code (subtypes not separately removed) could shrink it further. It should be read as a permutation-validated floor on within-line personalization at this scale, not as a usable per-line predictor. Second, all evaluation here is internal, leave-cell-line-out cross-validation within harmonized cell-line data; it does not establish transfer to independent screens, patient-derived models, or clinical response, and cross-study transfer is known to be substantially harder than within-study evaluation [13, 15]. Third, per-disease panels are small (20–70 lines) against tens of thousands of features, which fundamentally limits per-tissue personalization and inflates the variance of any biomarker estimate; the permutation false-discovery proportion among the 971 surviving hits is approximately 0.4 (≈ 392 expected by chance at $p < 0.05$), so individual hits require corroboration. Fourth, `sens_z` is a relative within-drug score, so only within-drug ordering, not absolute potency, is comparable across cohorts. Finally, cell lines are imperfect surrogates for tumours, and gene-level copy number was not incorporated in the present feature set.

4.5 Future directions

Four extensions follow directly from what the benchmark establishes. First, the target itself has moved. Because the pan-cancer signal survives removal of tissue of origin but is dwarfed by it (Section 4.1), feature sets should henceforth be judged against the *tissue-free* residual, where any real personalization must live, rather than against the raw drug-residualized target, on which tissue structure masks whether a feature adds anything at all. Second, now that the per-drug sweep spans every eligible panel, the natural next step is formal false-discovery control across the full set of 7,836 rather than the per-panel rate reported here, together with extension of the same sweep to further screens. Third, on the feature side, the finding that raw genetics dilute the estimator motivates feature-selected or pathway-prior genetic features, and drug-side features (target class or structural fingerprints) that let the model share information across chemically or mechanistically similar compounds. Fourth, translational credibility requires testing the mechanism-specific tier for transfer, against an independent screen such as a repurposing library, against patient-derived xenografts, or by imputation into patient cohorts, coupled to matching on pathway-level signatures, which are expected to be less sensitive to cross-platform batch effects than raw expression though that expectation is itself untested here; the *Semblance* engine (Section 2.8) is the mechanism of that rank-based matching. To support these extensions, the harmonized resource, the baseline-anchored evaluation harness, the permutation-validated signature atlas (whose per-hit pathway signatures are

illustrated in *Figure S4*), the semantic-matching engine, and the deployed application are released openly.

5. Conclusion

Harmonizing two major drug-sensitivity screens with multi-omic cell-line data and evaluating prediction honestly (against a drug-mean baseline, under leave-cell-line-out cross-validation, with permutation-controlled discovery) shows that the most apparent predictability comes from drug main effects and tissue of origin rather than molecular personalization, only a small within-line signal survives once tissue of origin is removed from the target. A two-stage significance protocol, a permutation null followed by independent corroboration, was applied across all 7,836 drug-tissue panels and retained permutation-validated, mechanism-specific biomarkers for a well-powered minority of agents, recovering known pharmacology and surfacing recurrent novel vulnerabilities. The contribution of this work is a reusable, structure-harmonized GDSC/CTRP-CCLE resource together with an evaluation discipline (baseline-anchored, permutation-controlled, and confidence-tiered) for drug-sensitivity prediction.

Statements and Declarations

Funding. The author received no funding for this work. **Competing interests.** The author declares no competing interests. **Ethics approval / consent.** Not applicable; the study uses publicly available cancer cell-line data only. **Data availability.** All primary data are public: GDSC (<https://www.cancerrxgene.org>), CTRP/CCLE via the Broad Institute, and CCLE/DepMap (<https://depmap.org>). The harmonized derived stores are archived together with the analysis code in a single Zenodo deposit, <https://doi.org/10.5281/zenodo.21764627>. **Code availability.** Analysis code (harmonization, modeling and evaluation harness, and the distributed batch fan-out) is available at <https://github.com/YueYvetteHao/Semblance2.0> and archived in the same Zenodo deposit as the data, <https://doi.org/10.5281/zenodo.21764627>. The accompanying confidence-tiered recommendation application, *Semblance 2.0*, is hosted at <https://yueyvettehao.github.io/Semblance2.0/>. The *Semblance* semantic-matching engine supporting pairwise gene-set-enrichment comparison is released separately as an open-source tool, *Semblance 1.0*, at <https://huggingface.co/spaces/yueyvettehao/Semblance>. **Author contributions.** Y.H. is the sole author and conceived the study, performed all analyses, and wrote the manuscript. **Use of AI tools.** AI-assisted coding and drafting were used; all analyses, results, and conclusions were verified by the author, who takes full responsibility for the content.

References

1. Garnett MJ, Edelman EJ, Heidorn SJ, et al (2012) Systematic identification of genomic markers of drug sensitivity in cancer cells. *Nature* 483(7391):570–575. <https://doi.org/10.1038/nature11005>
2. Yang W, Soares J, Greninger P, et al (2013) Genomics of Drug Sensitivity in Cancer (GDSC): a resource for therapeutic biomarker discovery in cancer cells. *Nucleic Acids Res* 41(D1):D955–D961. <https://doi.org/10.1093/nar/gks1111>

3. Iorio F, Knijnenburg TA, Vis DJ, et al (2016) A landscape of pharmacogenomic interactions in cancer. *Cell* 166(3):740–754. <https://doi.org/10.1016/j.cell.2016.06.017>
4. Basu A, Bodycombe NE, Cheah JH, et al (2013) An interactive resource to identify cancer genetic and lineage dependencies targeted by small molecules. *Cell* 154(5):1151–1161. <https://doi.org/10.1016/j.cell.2013.08.003>
5. Seashore-Ludlow B, Rees MG, Cheah JH, et al (2015) Harnessing connectivity in a large-scale small-molecule sensitivity dataset. *Cancer Discov* 5(11):1210–1223. <https://doi.org/10.1158/2159-8290.CD-15-0235>
6. Rees MG, Seashore-Ludlow B, Cheah JH, et al (2016) Correlating chemical sensitivity and basal gene expression reveals mechanism of action. *Nat Chem Biol* 12(2):109–116. <https://doi.org/10.1038/nchembio.1986>
7. Barretina J, Caponigro G, Stransky N, et al (2012) The Cancer Cell Line Encyclopedia enables predictive modeling of anticancer drug sensitivity. *Nature* 483(7391):603–607. <https://doi.org/10.1038/nature11003>
8. Ghandi M, Huang FW, Jané-Valbuena J, et al (2019) Next-generation characterization of the Cancer Cell Line Encyclopedia. *Nature* 569(7757):503–508. <https://doi.org/10.1038/s41586-019-1186-3>
9. Tsherniak A, Vazquez F, Montgomery PG, et al (2017) Defining a cancer dependency map. *Cell* 170(3):564–576.e16. <https://doi.org/10.1016/j.cell.2017.06.010>
10. Costello JC, Heiser LM, Georgii E, et al (2014) A community effort to assess and improve drug sensitivity prediction algorithms. *Nat Biotechnol* 32(12):1202–1212. <https://doi.org/10.1038/nbt.2877>
11. Adam G, Rampášek L, Safikhani Z, et al (2020) Machine learning approaches to drug response prediction: challenges and recent progress. *npj Precis Oncol* 4:19. <https://doi.org/10.1038/s41698-020-0122-1>
12. Baptista D, Ferreira PG, Rocha M (2021) Deep learning for drug response prediction in cancer. *Brief Bioinform* 22(1):360–379. <https://doi.org/10.1093/bib/bbz171>
13. Xia F, Allen J, Balaprakash P, et al (2022) A cross-study analysis of drug response prediction in cancer cell lines. *Brief Bioinform* 23(1):bbab356. <https://doi.org/10.1093/bib/bbab356>
14. Jang IS, Chaibub Neto E, Guinney J, Friend SH, Margolin AA (2014) Systematic assessment of analytical methods for drug sensitivity prediction from cancer cell line data. *Pac Symp Biocomput* 2014:63–74. https://doi.org/10.1142/9789814583220_0007
15. Haibe-Kains B, El-Hachem N, Birkbak NJ, et al (2013) Inconsistency in large pharmacogenomic studies. *Nature* 504(7480):389–393. <https://doi.org/10.1038/nature12831>
16. Geeleher P, Cox NJ, Huang RS (2016) Cancer biomarker discovery is improved by accounting for variability in general levels of drug sensitivity in pre-clinical models. *Genome Biol* 17(1):190. <https://doi.org/10.1186/s13059-016-1050-9>
17. Geeleher P, Cox NJ, Huang RS (2014) Clinical drug response can be predicted using baseline gene expression levels and in vitro drug sensitivity in cell lines. *Genome Biol* 15(3):R47. <https://doi.org/10.1186/gb-2014-15-3-r47>
18. Heller SR, McNaught A, Pletnev I, Stein S, Tchekhovskoi D (2015) InChI, the IUPAC International Chemical Identifier. *J Cheminform* 7:23. <https://doi.org/10.1186/s13321-015-0068-4>
19. Landrum G, et al RDKit: Open-source cheminformatics. <https://www.rdkit.org>
20. Kim S, Chen J, Cheng T, et al (2023) PubChem 2023 update. *Nucleic Acids Res* 51(D1):D1373–D1380. <https://doi.org/10.1093/nar/gkac956>
21. Pedregosa F, Varoquaux G, Gramfort A, et al (2011) Scikit-learn: machine learning in Python. *J Mach Learn Res* 12:2825–2830. <http://jmlr.org/papers/v12/pedregosa11a.html>
22. Zou H, Hastie T (2005) Regularization and variable selection via the elastic net. *J R Stat Soc*

- Series B Stat Methodol 67(2):301–320. <https://doi.org/10.1111/j.1467-9868.2005.00503.x>
23. Subramanian A, Tamayo P, Mootha VK, et al (2005) Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *Proc Natl Acad Sci USA* 102(43):15545–15550. <https://doi.org/10.1073/pnas.0506580102>
 24. Barbie DA, Tamayo P, Boehm JS, et al (2009) Systematic RNA interference reveals that oncogenic KRAS-driven cancers require TBK1. *Nature* 462(7269):108–112. <https://doi.org/10.1038/nature08460>
 25. Hänzelmann S, Castelo R, Guinney J (2013) GSEA: gene set variation analysis for microarray and RNA-seq data. *BMC Bioinformatics* 14:7. <https://doi.org/10.1186/1471-2105-14-7>
 26. Jassal B, Matthews L, Viteri G, et al (2020) The Reactome pathway knowledgebase. *Nucleic Acids Res* 48(D1):D498–D503. <https://doi.org/10.1093/nar/gkz1031>
 27. Liberzon A, Birger C, Thorvaldsdóttir H, et al (2015) The Molecular Signatures Database (MSigDB) hallmark gene set collection. *Cell Syst* 1(6):417–425. <https://doi.org/10.1016/j.cels.2015.12.004>
 28. Korotkevich G, Sukhov V, Budin N, et al (2021) Fast gene set enrichment analysis. *bioRxiv*. <https://doi.org/10.1101/060012>
 29. Love MI, Huber W, Anders S (2014) Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biol* 15(12):550. <https://doi.org/10.1186/s13059-014-0550-8>
 30. Harris CR, Millman KJ, van der Walt SJ, et al (2020) Array programming with NumPy. *Nature* 585(7825):357–362. <https://doi.org/10.1038/s41586-020-2649-2>
 31. Virtanen P, Gommers R, Oliphant TE, et al (2020) SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nat Methods* 17(3):261–272. <https://doi.org/10.1038/s41592-019-0686-2>
 32. Argo Workflows Authors The Argo Project: Argo Workflows, the workflow engine for Kubernetes. Cloud Native Computing Foundation. <https://argoproj.github.io/workflows>
 33. Remy F, Demuynck K, Demeester T (2024) BioLORD-2023: semantic textual representations fusing large language models and clinical knowledge graph insights. *J Am Med Inform Assoc* 31(9):1844–1855. <https://doi.org/10.1093/jamia/ocae029>
 34. Conneau A, Lample G, Ranzato M, Denoyer L, Jégou H (2018) Word translation without parallel data. In: *International Conference on Learning Representations (ICLR)*. <https://doi.org/10.48550/arXiv.1710.04087>
 35. Tsai J, Lee JT, Wang W, et al (2008) Discovery of a selective inhibitor of oncogenic B-Raf kinase with potent antimelanoma activity. *Proc Natl Acad Sci USA* 105(8):3041–3046. <https://doi.org/10.1073/pnas.0711741105>
 36. Vassilev LT, Vu BT, Graves B, et al (2004) In vivo activation of the p53 pathway by small-molecule antagonists of MDM2. *Science* 303(5659):844–848. <https://doi.org/10.1126/science.1092472>
 37. Viswanathan VS, Ryan MJ, Dhruv HD, et al (2017) Dependency of a therapy-resistant state of cancer cells on a lipid peroxidase pathway. *Nature* 547(7664):453–457. <https://doi.org/10.1038/nature23007>
 38. Tsoi J, Robert L, Paraiso K, et al (2018) Multi-stage differentiation defines melanoma subtypes with differential vulnerability to drug-induced iron-dependent oxidative stress. *Cancer Cell* 33(5):890–904.e5. <https://doi.org/10.1016/j.ccell.2018.03.017>

Supplementary Tables

Table S1. Prospective recovery of established drug–target pharmacology

Every screened combination of seven drug classes with the tissues in which that class’s target is an established dependency. The reference set was fixed from target biology before any result was consulted, so it is not the set of pairs quoted in Section 3.5, which were selected after validating. 28 of 53 panels validate (53%) against a background rate of 12.4% across all 7,836 panels (4.3-fold, one-sided binomial $p = 1.5\text{e-}12$). A panel counts as validated when its permutation $p < 0.05$; every failing panel is listed so the misses are visible.

Class	Validated Misses (drug, tissue, permutation p)	
BCL2	1 / 6	Venetoclax (DLBC, 0.13); navitoclax (LAML, 0.21); Navitoclax (DLBC, 0.35); Venetoclax (LAML, 0.36); navitoclax (ALL, 0.38)
BCR-ABL/SRC	0 / 3	dasatinib (ALL, 0.39); nilotinib (ALL, 0.61); dasatinib (LAML, 0.93)
BRAF	5 / 6	PLX-4032 (COAD/READ, 0.45)
EGFR/HER2	7 / 12	gefitinib (HNSC, 0.06); gefitinib (LUAD, 0.07); erlotinib (HNSC, 0.10); erlotinib (LUAD, 0.11); afatinib (HNSC, 0.30)
IGF1R	6 / 9	NVP-ADW742 (BRCA, 0.10); linsitinib (BRCA, 0.36); BMS-754807 (OV, 0.61)
MDM2/p53	5 / 5	none
MEK	4 / 12	trametinib (COAD/READ, 0.07); selumetinib (COAD/READ, 0.07); selumetinib (LUAD, 0.08); Refametinib (COAD/READ, 0.11); Refametinib (SKCM, 0.28); trametinib (PAAD, 0.40); Refametinib (PAAD, 0.62); selumetinib (PAAD, 0.71)

Total: 28 / 53 (53%) versus 12.4% background.

Supplementary Figures

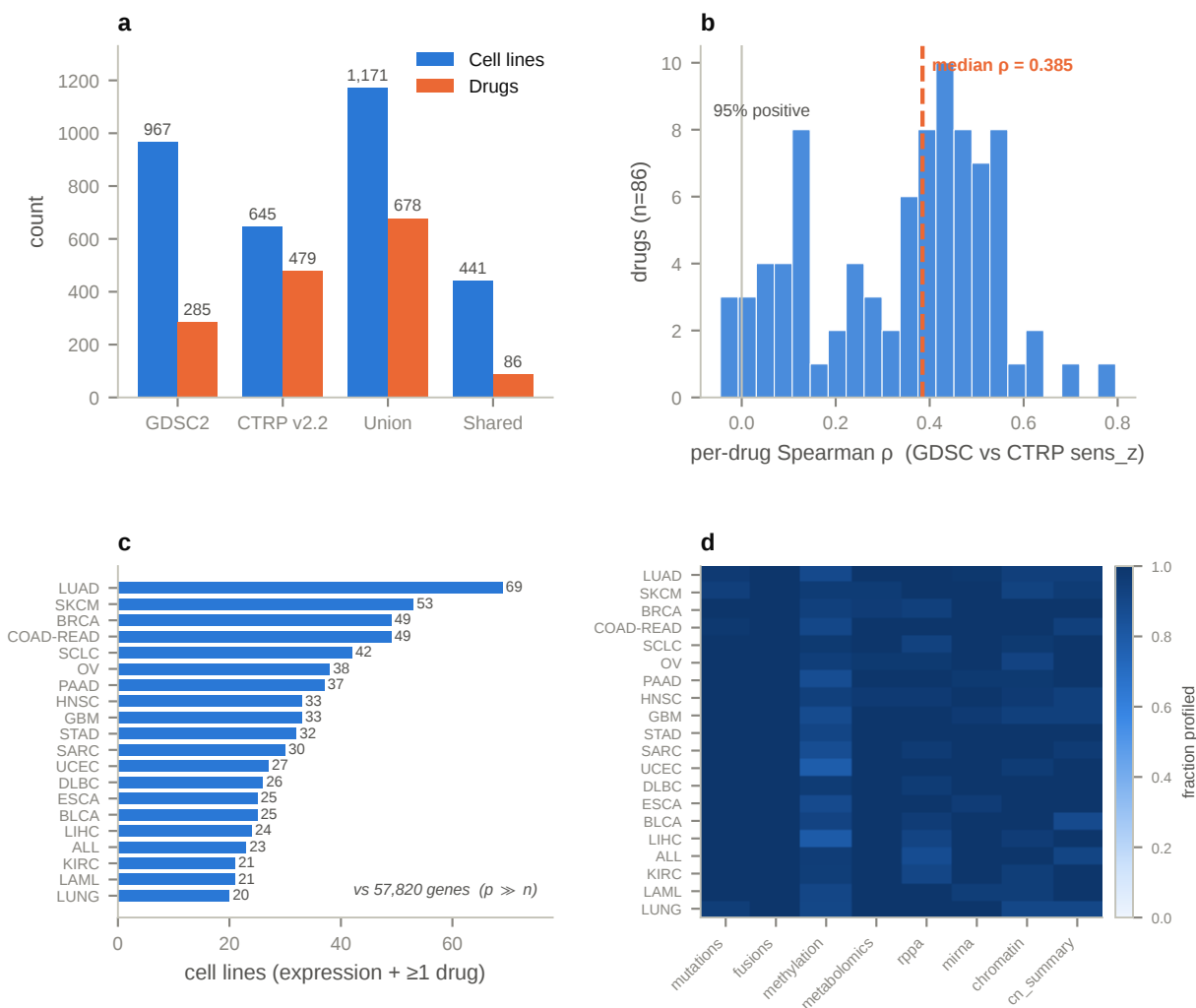


Figure S1. A structure-resolved GDSCUCTRP resource joined to CCLE multi-omics. (a) Cell lines and compounds contributed by each screen, their union, and the structure-confirmed shared set. Resolving drug identity on the InChIKey connectivity block rather than on compound name is what raises the shared-compound count from 71 by name to the 86 shown here. (b) Cross-cohort per-drug concordance of the harmonized metric (median Spearman $\rho = 0.385$). (c) Cell lines per disease against 57,820 genes, the $p \gg n$ regime that constrains modeling. (d) Multi-omic coverage by disease across the eight companion layers.

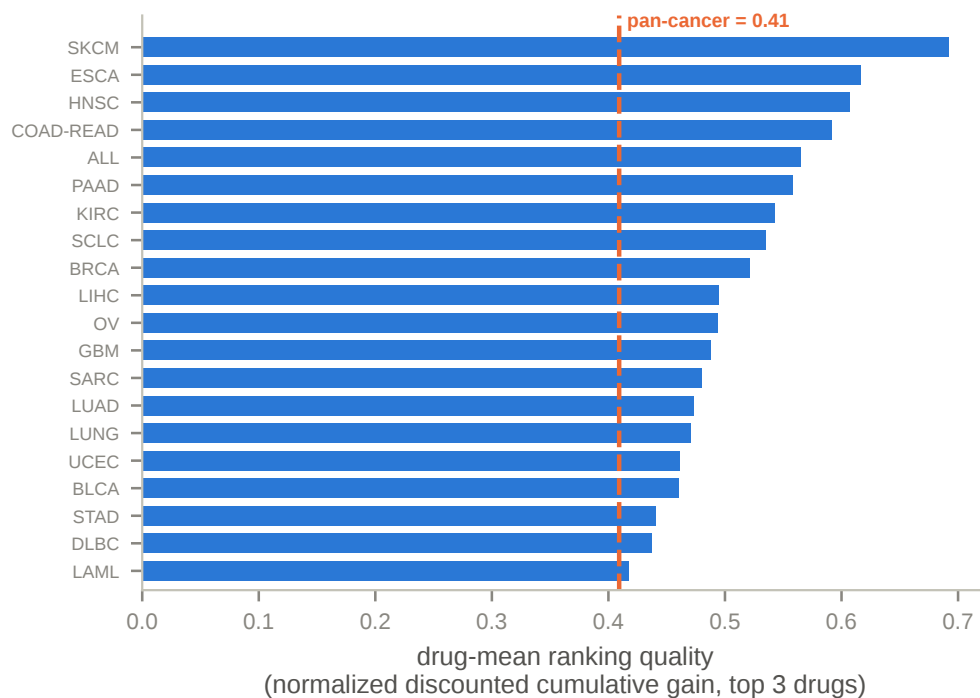


Figure S2. Ranking quality of the broad-spectrum drug-mean model by disease, measured as normalized discounted cumulative gain over the three top-ranked drugs, which scores how sensitive a cell line actually is to the drugs the model places first. Within-tissue ranking is informative, but the pan-cancer pooled value (dashed line, 0.41) is indistinguishable from the 0.416 expected when each cell line's drugs are ranked at random, reinforcing that broad-spectrum potency is a within-tissue property.

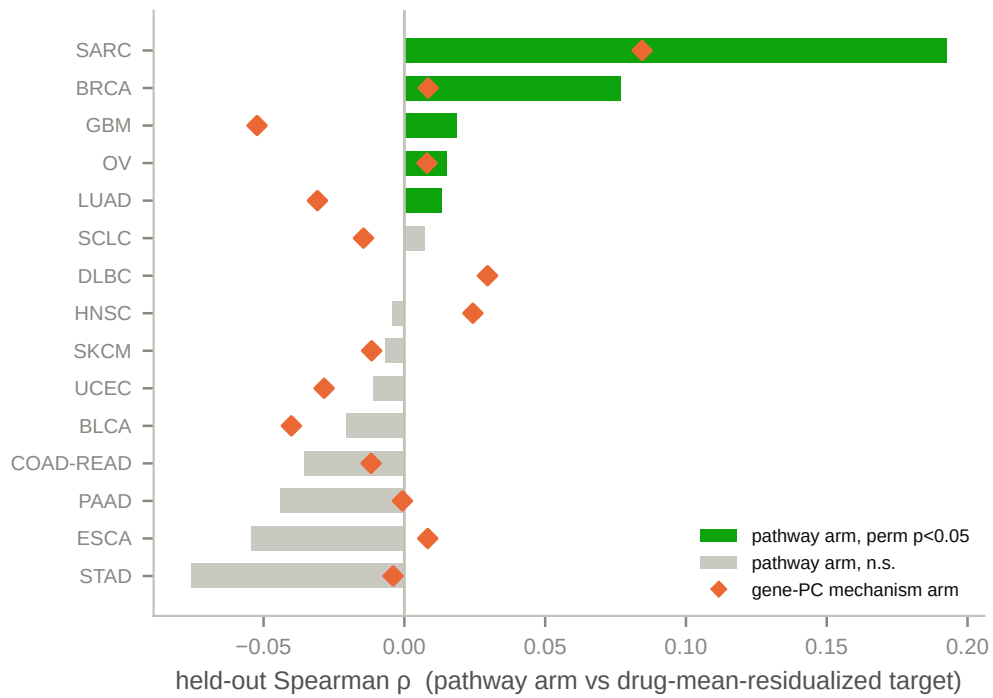


Figure S3. Pathway re-representation does not unlock personalization. Held-out Spearman of the raw ~1,800-feature Reactome pathway arm across the 15 disease panels that support the ssGSEA layer (bars coloured by permutation significance), with the gene-PC mechanism arm shown as reference diamonds; only sarcoma (+0.19) and breast (+0.08) exceed $\rho = 0.05$.

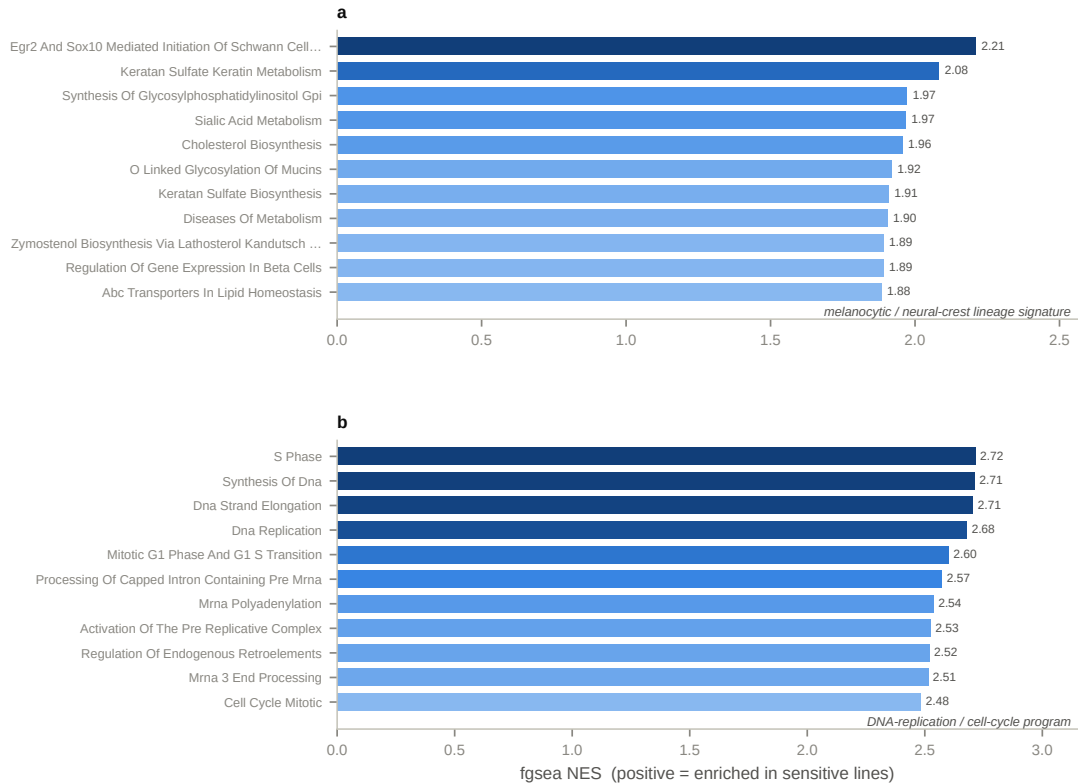


Figure S4. Sensitivity-associated Reactome programs for two permutation-validated hits, from the pre-ranked fgsea atlas (oriented so that positive normalized enrichment score marks programs enriched in sensitive lines); each panel shows the top programs by normalized enrichment score, with the value annotated at each bar. (a) PLX-4720 (BRAF inhibitor) in melanoma (held-out $\rho = 0.50$, permutation $p = 0.0099$, $n = 53$) is dominated by melanocytic and neural-crest lineage programs, a validated within-tissue signature that is nonetheless lineage-linked rather than target-specific. (b) The IGF1R inhibitor linsitinib in sarcoma (held-out $\rho = 0.43$, permutation $p = 0.030$, $n = 30$) shows a coherent DNA-replication and cell-cycle program, consistent with growth-factor-driven proliferation in the sensitive lines.